

**Geometric-Comb Interpolation, Gaussian Pascal
Transforms,
and the Fabius–Rvachev Boundary Layer**
**Lagrange/Newton theory on $cq^{\mathbb{N}_0}$, exact q -residuals,
a sharp stability dichotomy, and dyadic atomic-function applications**

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Abstract

Let $0 < q < 1$, $c > 0$, and consider the finite geometric comb $\{c, cq, \dots, cq^n\}$, together with its accumulation point 0. At these nodes, ordinary Newton interpolation is already a piece of q -calculus: its basis polynomials are the shifted factorials

$$N_k(x; c, q) = \prod_{r=0}^{k-1} (x - cq^r) = (-c)^k q^{\binom{k}{2}} (x/c; q^{-1})_k,$$

and its divided differences are normalized iterates of the Jackson q -derivative. This report develops that observation into a systematic interpolation theory. It proves explicit Lagrange and barycentric formulas, a Gaussian-Pascal change of basis between monomials and geometric Newton polynomials, an exact complete-homogeneous residual formula for every monomial and analytic function, a convergent infinite Newton expansion for functions analytic across the accumulation point, and an exact Newton–Jackson quadrature identity.

The stability theory has a particularly sharp two-sided answer. Extrapolation to the accumulation point is uniformly bounded:

$$\Lambda_n(0) = \frac{(-q; q)_n}{(q; q)_n} \longrightarrow \frac{(-q; q)_\infty}{(q; q)_\infty}.$$

In contrast, the global Lebesgue constant on $[0, c]$ is superexponential. A boundary-layer analysis proved here gives

$$\max_{0 \leq x \leq c} \Lambda_n(x) \sim \frac{2(-q; q)_\infty}{e} \frac{q^{-\binom{n}{2}}}{n}, \quad \frac{x_n^*}{c} = 1 - \frac{1}{n} + o(n^{-1}).$$

Thus the same comb is a stable extrapolation device at the end where it accumulates and a violently ill-conditioned global interpolation device in the persistent outer gap.

For $q = 1/2$, the nodes are precisely the inverse-dyadic scales naturally selected by the Fabius function. Writing $a_j = \mathbb{E}X^j/j!$ for the Fabius moment coefficients, the geometric Newton coefficients become the exact base-2 Gaussian transform

$$[F; 1, 2^{-1}, \dots, 2^{-k}] = \frac{1}{(1/2; 1/2)_k} \sum_{j=0}^k (-1)^j \begin{bmatrix} k \\ j \end{bmatrix}_2 a_j.$$

The report derives a finite Gaussian-cardinal formula for the entire interpolant, a q -Borel-type generating function for its boundary residues, and superexponential bounds proving recovery of every fixed flat endpoint jet even though F is nonanalytic there. Simultaneously, exact rational experiments show rapid divergence inside the gap $(1/2, 1)$ and motivate a sharp saddle-point conjecture. Reflecting $x = 1 - t$ transfers the results to the Rvachev function $\text{up}(t) = F(1 - |t|)$ on its boundary comb $t_j = 1 - 2^{-j}$. Finally, exact

polynomial reproduction by translates of ψ is composed with the geometric Lagrange operator, producing a finite Gaussian–Appell decoder and a closed interpolation/synthesis loop.

Statements labeled theorem, proposition, lemma, or corollary are proved. Unproved asymptotic claims are explicitly labeled conjectures. The bundled Python program uses exact rational arithmetic wherever possible and logarithmic floating-point arithmetic only for the enormous Lebesgue constants.

Executive summary of the main formulas

The geometric comb is more than a convenient nonuniform grid. It is the node geometry for which the Newton product, divided-difference operator, Gaussian coefficients, complete homogeneous functions, and Jackson integration all close under explicit finite formulae.

For $x_j = cq^j$ and $N_k(x) = \prod_{r < k} (x - x_r)$,

$$\begin{aligned} N_k(x) &= (-c)^k q^{\binom{k}{2}} (x/c; q^{-1})_k, \\ [f; x_0, \dots, x_k] &= \frac{D_q^k f(c)}{[k]_q!} \\ &= c^{-k} \sum_{j=0}^k \frac{(-1)^j q^{-jk + \binom{j+1}{2}}}{(q; q)_j (q; q)_{k-j}} f(cq^j), \\ x^m &= \sum_{k=0}^m c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q N_k(x), \\ N_k(x) &= \sum_{m=0}^k (-1)^{k-m} c^{k-m} q^{\binom{k-m}{2}} \begin{bmatrix} k \\ m \end{bmatrix}_q x^m. \end{aligned}$$

The last two displays are mutually inverse Gaussian-Pascal transforms.

For the degree- n interpolant $\mathcal{I}_n f$,

$$x^{n+1+r} - \mathcal{I}_n(x^{n+1+r}) = N_{n+1}(x) h_r(x, c, cq, \dots, cq^n),$$

and

$$h_r(x, c, cq, \dots, cq^n) = \sum_{s=0}^r x^s c^{r-s} \begin{bmatrix} n+r-s \\ n \end{bmatrix}_q.$$

This is the exact all-orders interpolation residual; the usual remainder with an unspecified intermediate derivative is replaced by an explicit q -polynomial.

For the Fabius function and $q = 1/2$,

$$\begin{aligned}
 F(2^{-j}) &= 2^{-\binom{j}{2}} a_j, \\
 \mathcal{I}_n F(x) &= \frac{\prod_{r=0}^n (x - 2^{-r})}{(1/2; 1/2)_n} \sum_{j=0}^n \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_2 a_j}{x - 2^{-j}}, \\
 \sum_{n \geq 0} (\mathcal{I}_n F)(0) z^n &= (z/2; 1/2)_\infty \sum_{j \geq 0} \frac{2^{-\binom{j}{2}} a_j}{(1/2; 1/2)_j} z^j.
 \end{aligned}$$

At $x = 0$ the residues decay at least like $2^{-\lfloor n^2/4 \rfloor}$ up to a fixed factor; at $x = 3/4$ they grow, according to exact rational data, on the conjectural scale $2^{n^2/4} n^{-n/2} e^{O(n)}$.

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Part I

Algebra of the geometric comb

Scope, provenance, and normalization

1.1 Relation to the source monograph and neighboring reports

The starting point is the chapter on negative upper indices and geometric Newton interpolation in `q_pochhammer_q_binomial_monograph.tex`. That chapter already records the basic divided-difference formula on $1, q, \dots, q^n$. The present report expands that seed in four directions:

1. it develops the full change-of-basis and residual algebra, rather than treating geometric interpolation as an isolated application;
2. it separates local convergence near the accumulation point from global stability on the convex hull and proves sharp asymptotics for both;
3. it specializes the transforms to the inverse-dyadic Fabius data and obtains exact boundary-residue and endpoint-jet estimates;
4. it composes the geometric interpolant with the existing Rvachev polynomial-synthesis operator, thereby connecting q -Pochhammer coordinates to finite dictionaries of compactly supported atomic functions.

The uniform dyadic-comb report is used only for comparison: its nodes $j/2^m$ fill $[0, 1]$ but suffer the familiar global high-degree instability. The geometric comb 2^{-j} has a different pathology: it samples one boundary on all binary scales, but it leaves a fixed gap $(1/2, 1)$. The Rvachev–Lagrange loop report supplies the exact inverse-moment Appell synthesis of polynomials by translates of up. Results imported from those reports are identified when used; the new derivations are proved here.

All novelty statements are relative to the audited repository snapshot in the title, not claims of unrestricted world priority. Classical background on q -series and q -calculus is available in [1, 7, 9, 5]; the q -Newton viewpoint is particularly explicit in [10]. Barycentric interpolation and its numerical evaluation are treated in [4, 8].

1.2 The affine geometric comb

Definition 1.1 (Affine geometric comb). Let $\alpha, \beta, q \in \mathbb{C}$ with $\beta \neq 0$. The degree- n affine geometric comb is

$$\mathcal{G}_n(\alpha, \beta; q) := \{\alpha + \beta q^j : 0 \leq j \leq n\}.$$

When $|q| < 1$, its infinite continuation accumulates at α . The multiplicative normalization is $\alpha = 0$, $\beta = c$; the reflected boundary normalization used for up is $\alpha = 1$, $\beta = -1$.

Finite interpolation requires distinct nodes, equivalently $q^d \neq 1$ for $1 \leq d \leq n$. Most algebraic formulae remain valid over an arbitrary field under that hypothesis. The convergence and Lebesgue theory will assume

$$0 < q < 1, \quad c > 0, \quad x_j = cq^j. \tag{1.1}$$

The affine case is recovered by $z = (x - \alpha)/\beta$.

Proposition 1.2 (Affine covariance). *Let P interpolate f on $\mathcal{G}_n(\alpha, \beta; q)$ and set $g(z) = f(\alpha + \beta z)$. If Q interpolates g on $1, q, \dots, q^n$, then*

$$P(x) = Q\left(\frac{x - \alpha}{\beta}\right).$$

The Lagrange cardinals transform by the same substitution, and the Lebesgue function is invariant under real affine changes that map the interpolation interval bijectively to itself.

Proof. Both sides are polynomials of degree at most n and agree at all $n + 1$ nodes. The cardinal and Lebesgue assertions follow immediately. $\square$

Thus c is retained in algebraic formulae to expose homogeneity, but it can be set to 1 in all conditioning arguments.

1.3 Notation for q -analogues

For $k \in \mathbb{N}_0$,

$$(a; q)_k := \prod_{r=0}^{k-1} (1 - aq^r), \quad (a; q)_0 := 1, \quad (1.2)$$

$$[k]_q := \frac{1 - q^k}{1 - q}, \quad [k]_q! := \prod_{r=1}^k [r]_q = \frac{(q; q)_k}{(1 - q)^k}, \quad (1.3)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} \quad (0 \leq k \leq n). \quad (1.4)$$

The Gaussian coefficient is a polynomial in q ; quotient notation is used only where the displayed denominators are nonzero. Reciprocity reads

$$\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (1.5)$$

The complete homogeneous symmetric polynomial is characterized by

$$\prod_{j=0}^m \frac{1}{1 - y_j t} = \sum_{r \geq 0} h_r(y_0, \dots, y_m) t^r. \quad (1.6)$$

Its geometric principal specialization is

$$h_r(c, cq, \dots, cq^m) = c^r \begin{bmatrix} m + r \\ m \end{bmatrix}_q. \quad (1.7)$$

This simple identity is the source of almost every Gaussian coefficient in the residual formulae below.

Newton interpolation as q -Taylor calculus

2.1 The q -Pochhammer Newton basis

Definition 2.1 (Geometric Newton polynomial). For $k \geq 0$, define

$$N_k(x; c, q) := \prod_{r=0}^{k-1} (x - cq^r), \quad N_0 := 1. \quad (2.1)$$

The factor locations are exactly the first k nodes of the comb. Pulling out $-cq^r$ from each factor gives the fundamental identification

$$N_k(x; c, q) = (-c)^k q^{\binom{k}{2}} (x/c; q^{-1})_k. \quad (2.2)$$

Thus the finite q -Pochhammer symbol is not merely analogous to a Newton basis: on a geometric grid it *is* the Newton basis, up to its monic normalization.

Let

$$(D_q f)(x) := \frac{f(x) - f(qx)}{(1-q)x}, \quad x \neq 0, \quad (2.3)$$

with the removable value at 0 used when it exists. The usual q -Leibniz rule gives

$$D_q N_k(x; c, q) = [k]_q N_{k-1}(x; c, q). \quad (2.4)$$

A direct proof follows by writing $N_k = (x - c)_q^k$ in the standard q -falling-power notation. The next theorem supplies the exact data-side operator and fixes every power of q .

Theorem 2.2 (Iterated Jackson difference and geometric divided difference). Let $E_q f(x) = f(qx)$. For every $k \geq 0$,

$$D_q^k = \frac{x^{-k}}{(1-q)^k} (E_q; q^{-1})_k, \quad (2.5)$$

$$D_q^k f(x) = \frac{x^{-k}}{(1-q)^k} \sum_{j=0}^k (-1)^j q^{-jk + \binom{j+1}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q f(q^j x). \quad (2.6)$$

Consequently, for distinct nodes $c, cq, \dots, cq^k$,

$$[f; c, cq, \dots, cq^k] = \frac{D_q^k f(c)}{[k]_q!} = c^{-k} \sum_{j=0}^k \frac{(-1)^j q^{-jk + \binom{j+1}{2}}}{(q; q)_j (q; q)_{k-j}} f(cq^j). \quad (2.7)$$

Proof. As operators,

$$D_q = \frac{x^{-1}(1 - E_q)}{1 - q}.$$

The commutation relation $E_q x^{-1} = q^{-1} x^{-1} E_q$ implies inductively

$$D_q^k = \frac{x^{-k}}{(1 - q)^k} \prod_{r=0}^{k-1} (1 - q^{-r} E_q),$$

which is (2.5). The finite q -binomial theorem at base q^{-1} gives

$$(E_q; q^{-1})_k = \sum_{j=0}^k (-1)^j q^{-\binom{j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_{q^{-1}} E_q^j.$$

Using reciprocity (1.5), the exponent becomes $-\binom{j}{2} - j(k - j) = -jk + \binom{j+1}{2}$, proving (2.6).

It remains to identify the normalization. Both functionals $[f; c, cq, \dots, cq^k]$ and $D_q^k f(c)/[k]_q!$ vanish on polynomials of degree less than k . Applied to x^k , the divided difference is 1, while $D_q^k x^k = [k]_q!$. Equality follows on polynomials of degree at most k ; the explicit divided-difference formula, being a fixed linear combination of point evaluations, then proves the equality for arbitrary data on the nodes. Substituting $x = c$ in (2.6) and using $(1 - q)^k [k]_q! = (q; q)_k$ gives the last expression. $\square$

Corollary 2.3 (q -Taylor form of the interpolant). *The polynomial of degree at most n interpolating f at $c, cq, \dots, cq^n$ is*

$$(\mathcal{I}_n f)(x) = \sum_{k=0}^n \frac{D_q^k f(c)}{[k]_q!} N_k(x; c, q). \quad (2.8)$$

Proof. This is the ordinary Newton interpolation formula with Theorem 2.2 substituted for each divided difference. $\square$

Remark 2.4 (Two meanings of “ q -Taylor”). Formula (2.8) is simultaneously a polynomial q -Taylor formula and a finite interpolation formula. No limit process is involved. Infinite q -Taylor series require convergence hypotheses; those are developed in Chapter 6.

2.2 Triangular evaluation and the geometric Vandermonde determinant

At a node $x = cq^j$, the basis evaluation is triangular:

$$N_k(cq^j; c, q) = \begin{cases} (-c)^k q^{\binom{k}{2}} \frac{(q; q)_j}{(q; q)_{j-k}}, & k \leq j, \\ 0, & k > j. \end{cases} \quad (2.9)$$

This gives an $O(n^2)$ forward-substitution algorithm for the coefficients, but it also exposes the determinant exactly.

Theorem 2.5 (Geometric Vandermonde determinant). *Let $V_n = (x_j^m)_{0 \leq j, m \leq n}$ with $x_j = cq^j$, and put $N = n(n+1)/2$. Then*

$$\det V_n = (-1)^N c^N q^{\binom{n+1}{3}} \prod_{k=1}^n (q; q)_k. \quad (2.10)$$

Proof. The Vandermonde product is

$$\det V_n = \prod_{0 \leq r < s \leq n} (cq^s - cq^r).$$

For fixed s ,

$$\prod_{r=0}^{s-1} (q^s - q^r) = (-1)^s q^{\binom{s}{2}} (q; q)_s.$$

Multiplying over $s = 1, \dots, n$ gives the sign $(-1)^{\sum s} = (-1)^N$, the power $q^{\sum s \binom{s}{2}} = q^{\binom{n+1}{3}}$, and the stated product. There are N pairwise differences, giving c^N . $\square$

The factor $q^{\binom{n+1}{3}}$ already signals severe monomial-Vandermonde conditioning. Newton coordinates remove the linear solve, but they do not remove the intrinsic amplification of the interpolation operator. That amplification is measured exactly in [Chapter 7](#).

2.3 Online extension of the comb

Adding the new node cq^{n+1} does not change any earlier Newton coefficient. Only

$$\gamma_{n+1} = \frac{f(cq^{n+1}) - \sum_{k=0}^n \gamma_k N_k(cq^{n+1})}{N_{n+1}(cq^{n+1})} \quad (2.11)$$

is appended. Since the denominator is $(-c)^{n+1} q^{\binom{n+1}{2}} (q; q)_{n+1}$, the update is explicit and exact in any coefficient field containing c and q . This nesting is valuable for Fabius data: every additional sample resolves one more dyadic boundary scale without invalidating the existing expansion.

The Gaussian Pascal change of coordinates

3.1 Monomials versus geometric falling powers

Theorem 3.1 (Gaussian Pascal basis pair). *For every $m, k \in \mathbb{N}_0$,*

$$x^m = \sum_{k=0}^m c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q N_k(x; c, q), \quad (3.1)$$

$$N_k(x; c, q) = \sum_{m=0}^k (-1)^{k-m} c^{k-m} q^{\binom{k-m}{2}} \begin{bmatrix} k \\ m \end{bmatrix}_q x^m. \quad (3.2)$$

The two triangular coefficient arrays are mutually inverse.

Proof. The second identity is the finite q -binomial theorem applied to

$$N_k(x; c, q) = x^k (c/x; q)_k.$$

For the first, apply the divided-difference functional to x^m . A standard identity for monomial divided differences gives

$$[x^m; c, cq, \dots, cq^k] = h_{m-k}(c, cq, \dots, cq^k) = c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q,$$

where the value is zero for $k > m$. Substitution in the Newton interpolation formula proves (3.1). Since both transformations are unit triangular, either identity also proves that they are inverses. $\square$

Example 3.2. The first rows are

$$\begin{aligned} x &= c + N_1, \\ x^2 &= c^2 + c(1+q)N_1 + N_2, \\ x^3 &= c^3 + c^2(1+q+q^2)N_1 + c(1+q+q^2)N_2 + N_3. \end{aligned}$$

At $q \rightarrow 1$, these become the familiar binomial transforms between powers and falling factorials after the logarithmic scaling described in [Section 3.5](#).

3.2 Coefficient transforms and Rogers–Szegő polynomials

Let

$$f(x) = \sum_{m=0}^M a_m x^m = \sum_{k=0}^M \gamma_k N_k(x; c, q). \quad (3.3)$$

Then [Theorem 3.1](#) gives

$$\gamma_k = \sum_{m=k}^M a_m c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q, \quad (3.4)$$

$$a_m = \sum_{k=m}^M \gamma_k (-1)^{k-m} c^{k-m} q^{\binom{k-m}{2}} \begin{bmatrix} k \\ m \end{bmatrix}_q. \quad (3.5)$$

These are finite q -binomial inversions with a grading by the scale c . The diagonal entries are one, so the determinant of the basis-change matrix is one even though individual off-diagonal entries can be large.

Define the Rogers–Szegő polynomial

$$H_m(z; q) := \sum_{k=0}^m \begin{bmatrix} m \\ k \end{bmatrix}_q z^k. \quad (3.6)$$

Then the ordinary generating polynomial of the Newton coefficients satisfies

$$\sum_{k=0}^M \gamma_k z^k = \sum_{m=0}^M a_m c^m H_m(z/c; q). \quad (3.7)$$

Thus geometric interpolation naturally converts monomial coefficient data through a Rogers–Szegő transform. This is one route from interpolation to the orthogonal-polynomial side of q -analysis.

3.3 A matrix factorization of geometric sampling

Let $A = (a_0, \dots, a_n)^T$, $\Gamma = (\gamma_0, \dots, \gamma_n)^T$, and $Y = (f(c), f(cq), \dots, f(cq^n))^T$. Introduce

$$T_{k,m} = \begin{cases} c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q, & k \leq m, \\ 0, & k > m, \end{cases}$$

$$L_{j,k} = \begin{cases} (-c)^k q^{\binom{k}{2}} \frac{(q; q)_j}{(q; q)_{j-k}}, & k \leq j, \\ 0, & k > j. \end{cases}$$

Then

$$\Gamma = TA, \quad Y = L\Gamma = LTA. \quad (3.8)$$

The geometric Vandermonde matrix is therefore factored into a unit upper Gaussian-Pascal matrix and an explicitly invertible lower Newton-evaluation matrix. Formula (2.7) is the closed form for L^{-1} .

This factorization is useful in three distinct senses.

1. It separates algebraic basis conversion from evaluation geometry.
2. It gives exact arithmetic algorithms over rational or algebraic q .
3. It identifies where instability resides: not in a mysterious dense solve, but in the growth of the explicit triangular factors and the interpolation projector they define.

3.4 Combinatorics at reciprocal integer base

When $q = b^{-1}$ with an integer $b \geq 2$, reciprocity rewrites many coefficients with Gaussian polynomials at base b :

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = b^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_b. \quad (3.9)$$

For prime-power b , $\begin{bmatrix} n \\ k \end{bmatrix}_b$ counts k -dimensional subspaces of $\mathbb{F}_b^n$. In particular, the half-base interpolation transforms for the Fabius function contain literal counts of binary subspaces, multiplied by the powers of two dictated by reciprocity. This is a more rigid connection than the generic statement that “Gaussian coefficients occur”: the entire monomial/Newton conversion matrix is a rescaled incidence-count array for finite vector spaces.

3.5 The classical limit in logarithmic coordinates

As $q \rightarrow 1^-$ the nodes cq^j collide at c , so a direct limit in the x -coordinate is confluent. The correct resolved coordinate is

$$x = cq^\xi. \quad (3.10)$$

For fixed k and ξ ,

$$\lim_{q \rightarrow 1^-} \frac{(-1)^k N_k(cq^\xi; c, q)}{c^k (1-q)^k} = \prod_{r=0}^{k-1} (\xi - r) = \xi^k. \quad (3.11)$$

Indeed, $(q^\xi - q^r)/(1-q) \rightarrow r - \xi$. Hence geometric Newton interpolation is a multiplicative deformation of ordinary Newton interpolation on the integer grid in the logarithmic coordinate. Simultaneously, $\begin{bmatrix} m \\ k \end{bmatrix}_q \rightarrow \binom{m}{k}$, so (3.1) tends to the ordinary binomial/falling-factorial basis transform after the same rescaling.

Lagrange cardinals and geometric filters

4.1 Closed barycentric weights

Let

$$\omega_{n+1}(x) := \prod_{r=0}^n (x - cq^r) = N_{n+1}(x; c, q). \quad (4.1)$$

The leading-form barycentric weight is $\lambda_{j,n} = 1/\omega'_{n+1}(cq^j)$. Splitting factors below and above the j th node gives the following exact expression.

Theorem 4.1 (Geometric barycentric weights and cardinals). *For $0 \leq j \leq n$,*

$$\lambda_{j,n} = c^{-n} \frac{(-1)^j q^{\binom{j+1}{2} - nj}}{(q; q)_j (q; q)_{n-j}}, \quad (4.2)$$

$$\ell_{j,n}(x) = c^{-n} \frac{(-1)^j q^{\binom{j+1}{2} - nj}}{(q; q)_j (q; q)_{n-j}} \frac{\omega_{n+1}(x)}{x - cq^j}. \quad (4.3)$$

Consequently,

$$(\mathcal{I}_n f)(x) = \omega_{n+1}(x) \sum_{j=0}^n \frac{\lambda_{j,n} f(cq^j)}{x - cq^j}, \quad (4.4)$$

with the exact-node value supplied by continuity.

Proof. For $r < j$, $cq^j - cq^r = -cq^r(1 - q^{j-r})$; for $r > j$, $cq^j - cq^r = cq^j(1 - q^{r-j})$. Therefore

$$\begin{aligned} \prod_{r \neq j} (cq^j - cq^r) &= (-1)^j c^n q^{\binom{j}{2} + j(n-j)} (q; q)_j (q; q)_{n-j} \\ &= (-1)^j c^n q^{nj - \binom{j+1}{2}} (q; q)_j (q; q)_{n-j}. \end{aligned}$$

Inversion proves (4.2); the other formulas are the standard cardinal identities. $\square$

The quotient form often used for numerical evaluation is

$$(\mathcal{I}_n f)(x) = \frac{\sum_{j=0}^n \frac{\lambda_{j,n}}{x - cq^j} f(cq^j)}{\sum_{j=0}^n \frac{\lambda_{j,n}}{x - cq^j}}. \quad (4.5)$$

In exact arithmetic (4.4) and (4.5) agree. In floating point, evaluation far from or outside the node cluster can make their stability behavior differ; see [8, 13]. The experiments bundled with this report use the product form for Lebesgue functions because it contains no alternating cancellation after absolute values are taken.

4.2 Cardinals in Newton coordinates

Proposition 4.2 (Triangular Newton expansion of a cardinal). *For each j ,*

$$\ell_{j,n}(x) = \sum_{k=j}^n c^{-k} \frac{(-1)^j q^{-jk + \binom{j+1}{2}}}{(q; q)_j (q; q)_{k-j}} N_k(x; c, q). \quad (4.6)$$

Proof. Apply (2.7) to the data sequence that is one at node j and zero elsewhere. The k th divided difference vanishes for $k < j$ and equals the displayed coefficient for $k \geq j$. $\square$

This formula makes the sample-to-Newton map manifestly lower triangular in the sample index. It is useful when the comb is extended online: the old cardinals change, but the data-to-Newton coefficient of order k still depends only on the first $k + 1$ samples.

4.3 Evaluation at the accumulation point: q -Richardson weights

Although 0 is not among the finite nodes, evaluating the interpolant there produces a classical geometric extrapolation filter.

Proposition 4.3 (Accumulation-point weights). *Define $\mu_{j,n} := \ell_{j,n}(0)$. Then*

$$\mu_{j,n} = \frac{(-1)^{n-j} q^{\binom{n-j+1}{2}}}{(q; q)_j (q; q)_{n-j}}. \quad (4.7)$$

They satisfy

$$\sum_{j=0}^n \mu_{j,n} q^{jm} = \delta_{m0}, \quad 0 \leq m \leq n. \quad (4.8)$$

Thus $\sum_j \mu_{j,n} f(cq^j)$ removes the first n powers in a local expansion of f at zero.

Proof. Set $x = 0$ in (4.3) and simplify the powers of c and q . Equation (4.8) is polynomial reproduction applied to $1, x, \dots, x^n$ and evaluated at zero. $\square$

Theorem 4.4 (Exact Richardson tail). *Let $f(z) = \sum_{m \geq 0} a_m z^m$ be a formal power series. Then*

$$\sum_{j=0}^n \mu_{j,n} f(cq^j z) = a_0 + (-1)^n c^{n+1} q^{\binom{n+1}{2}} \sum_{r \geq 0} a_{n+1+r} c^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q z^{n+1+r}. \quad (4.9)$$

Whenever both sides converge analytically, the same identity holds as an analytic equality.

Proof. By (4.8), only powers $m \geq n + 1$ survive. The first surviving moment is

$$\sum_j \mu_{j,n} (cq^j)^{n+1} = (-1)^n c^{n+1} q^{\binom{n+1}{2}},$$

which is the negative of the nodal polynomial at zero. The universal moment residual theorem proved in Theorem 5.1, or a direct application of (1.7), gives the factor $c^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q$ for every higher moment. $\square$

4.4 Generating function of the Richardson triangle

For arbitrary sample data $y_0, y_1, \dots$, set

$$r_n := \sum_{j=0}^n \mu_{j,n} y_j. \quad (4.10)$$

Euler's expansion of an infinite q -Pochhammer product gives an exact transform identity.

Theorem 4.5 (Euler factorization of geometric extrapolation). *As a formal power series in z ,*

$$\boxed{\sum_{n \geq 0} r_n z^n = (qz; q)_\infty \sum_{j \geq 0} \frac{y_j}{(q; q)_j} z^j.} \quad (4.11)$$

Proof. Put $m = n - j$ in (4.7). The double sum factors as

$$\begin{aligned} \sum_{n \geq 0} r_n z^n &= \left(\sum_{m \geq 0} \frac{(-1)^m q^{\binom{m+1}{2}}}{(q; q)_m} z^m \right) \left(\sum_{j \geq 0} \frac{y_j}{(q; q)_j} z^j \right) \\ &= (qz; q)_\infty \sum_{j \geq 0} \frac{y_j}{(q; q)_j} z^j, \end{aligned}$$

where Euler's identity $(a; q)_\infty = \sum_{m \geq 0} (-1)^m q^{\binom{m}{2}} a^m / (q; q)_m$ was used with $a = qz$. $\square$

This factorization will become a q -Borel-type transform of the Fabius moment sequence in Section 10.1.

Part II

Exact residuals, convergence, and stability

Complete-homogeneous residual calculus

5.1 A universal moment theorem

The Gaussian residual is best understood as a specialization of a statement that contains no q -notation.

Theorem 5.1 (Universal moment residual). *Let $x_0, \dots, x_n$ and $w_0, \dots, w_n$ lie in a commutative ring, and put $M_m = \sum_{j=0}^n w_j x_j^m$. Suppose*

$$M_0 = \dots = M_{n-1} = 0, \quad M_n = K. \quad (5.1)$$

Then, for every $r \geq 0$,

$$M_{n+r} = K h_r(x_0, \dots, x_n). \quad (5.2)$$

Proof. In the formal power-series ring,

$$M(t) := \sum_{m \geq 0} M_m t^m = \sum_{j=0}^n \frac{w_j}{1 - x_j t}.$$

Let $E(t) = \prod_{j=0}^n (1 - x_j t)$. The product $E(t)M(t)$ is a polynomial of degree at most n . Its coefficient of t^m depends only on $M_0, \dots, M_m$. The hypotheses therefore make every coefficient below n vanish and make the coefficient at n equal K . Hence $E(t)M(t) = Kt^n$. Divide by $E(t)$ and use (1.6). $\square$

The theorem explains why complete homogeneous functions, rather than an ad hoc sequence of correction terms, govern all higher residual moments. On a geometric alphabet they become Gaussian coefficients by (1.7).

5.2 Exact monomial interpolation error

Theorem 5.2 (All-degree monomial residual). *Let $\mathcal{I}_n$ interpolate on $x_j = cq^j$, $0 \leq j \leq n$. For every $r \geq 0$,*

$$x^{n+1+r} - \mathcal{I}_n(x^{n+1+r}) = N_{n+1}(x; c, q) h_r(x, c, cq, \dots, cq^n). \quad (5.3)$$

Moreover,

$$h_r(x, c, cq, \dots, cq^n) = \sum_{s=0}^r x^s c^{r-s} \begin{bmatrix} n+r-s \\ n \end{bmatrix}_q. \quad (5.4)$$

Proof. The difference on the left is monic of degree $n+1+r$ and vanishes at every node, so it is divisible by $N_{n+1}(x)$. More precisely, the Newton remainder identity for the monomial t^{n+1+r} is

$$t^{n+1+r} - \mathcal{I}_n(t^{n+1+r}) = N_{n+1}(t)[x_0, \dots, x_n, t]z^{n+1+r}.$$

A monomial divided difference equals the complete homogeneous function of the nodes, giving (5.3). Alternatively, the quotient is symmetric and its generating function is immediate.

Indeed,

$$\begin{aligned} \sum_{r \geq 0} h_r(x, c, cq, \dots, cq^n) t^r &= \frac{1}{1 - xt} \frac{1}{(ct; q)_{n+1}} \\ &= \left(\sum_{s \geq 0} x^s t^s \right) \left(\sum_{u \geq 0} c^u \begin{bmatrix} n+u \\ n \end{bmatrix}_q t^u \right). \end{aligned}$$

Extracting the coefficient of t^r proves (5.4). $\square$

Example 5.3. The first three errors are

$$\begin{aligned} x^{n+1} - \mathcal{I}_n x^{n+1} &= N_{n+1}(x), \\ x^{n+2} - \mathcal{I}_n x^{n+2} &= N_{n+1}(x) (x + c[n+1]_q), \\ x^{n+3} - \mathcal{I}_n x^{n+3} &= N_{n+1}(x) \left(x^2 + c[n+1]_q x + c^2 \begin{bmatrix} n+2 \\ n \end{bmatrix}_q \right). \end{aligned}$$

The coefficients are positive for $0 < q < 1$, which is useful for one-sided error bounds when the Taylor coefficients of the target have a fixed sign.

5.3 Residual of a power series

Corollary 5.4 (Exact analytic residual kernel). *Suppose $f(x) = \sum_{m \geq 0} a_m x^m$ and the rearrangements below are absolutely convergent. Then*

$$f(x) - \mathcal{I}_n f(x) = N_{n+1}(x; c, q) \sum_{r \geq 0} a_{n+1+r} h_r(x, c, cq, \dots, cq^n) \quad (5.5)$$

$$= N_{n+1}(x; c, q) \sum_{r \geq 0} a_{n+1+r} \sum_{s=0}^r x^s c^{r-s} \begin{bmatrix} n+r-s \\ n \end{bmatrix}_q. \quad (5.6)$$

The same identity is valid coefficientwise for a formal power series.

Proof. Apply Theorem 5.2 term by term. In the formal setting every coefficient receives only finitely many contributions; in the analytic setting absolute convergence justifies the interchange. $\square$

Formula (5.6) is a geometric-grid counterpart of the Peano or Cauchy interpolation remainder, but it is exact and contains only the Taylor coefficients and Gaussian polynomials. It has three useful specializations:

1. $x = 0$ gives the Richardson tail (4.9);
2. $a_m = 0$ for $m > M$ makes the sum finite and exact over any coefficient ring;
3. coefficientwise positivity permits sharp majorants without unknown intermediate points.

5.4 Divided differences of analytic functions

The same algebra gives an explicit transform from Taylor coefficients to geometric divided differences.

Proposition 5.5 (Taylor-to-Newton transform). *If $f(x) = \sum_{m \geq 0} a_m x^m$ converges in a disk containing the nodes, then*

$$\gamma_k := [f; c, cq, \dots, cq^k] = \sum_{m \geq k} a_m c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q. \quad (5.7)$$

Proof. Apply linearity to the monomial divided difference used in the proof of [Theorem 3.1](#). Absolute convergence follows, for example, from the estimates in [Theorem 6.1](#). $\square$

This formula is often more useful than the nodal expression (2.7): the former exposes analytic growth, whereas the latter exposes exact arithmetic and sample dependence.

Infinite Newton series at an accumulating comb

6.1 Why an accumulation point changes the problem

For a generic infinite node sequence, matching all nodes need not determine an analytic function. Here the nodes cq^j accumulate at 0. If both the target and the Newton-series sum are analytic near 0, equality on the comb forces equality by the identity theorem. The main task is therefore to prove locally uniform convergence of the series

$$\sum_{k \geq 0} \gamma_k N_k(x; c, q). \quad (6.1)$$

Theorem 6.1 (Analytic convergence across the accumulation point). *Let $0 < q < 1$, $0 < c < R$, and suppose f is analytic in $|z| < R_0$ for some $R_0 > R$. Put*

$$M_R := \max_{|z|=R} |f(z)|.$$

For every $0 < r < R$, the geometric Newton coefficients satisfy

$$|\gamma_k| \leq \frac{M_R R^{-k}}{(q; q)_k (1 - c/R)}, \quad (6.2)$$

and the infinite Newton series converges absolutely and uniformly on $|x| \leq r$ to $f(x)$. Quantitatively,

$$\sup_{|x| \leq r} |f(x) - \mathcal{I}_n f(x)| \leq \frac{M_R (-c/r; q)_\infty}{(1 - c/R)(q; q)_\infty} \frac{(r/R)^{n+1}}{1 - r/R}. \quad (6.3)$$

For $r = 0$, convergence follows directly from the endpoint formula.

Proof. Cauchy's estimate gives $|a_m| \leq M_R R^{-m}$. By (5.7),

$$|\gamma_k| \leq \sum_{m \geq k} M_R R^{-m} c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q.$$

For $0 < q < 1$,

$$\begin{bmatrix} m \\ k \end{bmatrix}_q = \frac{(q^{m-k+1}; q)_k}{(q; q)_k} \leq \frac{1}{(q; q)_k}.$$

Summing the geometric majorant proves (6.2).

If $|x| \leq r$, then

$$|N_k(x; c, q)| \leq \prod_{j=0}^{k-1} (r + cq^j) = r^k (-c/r; q)_k \leq r^k (-c/r; q)_\infty.$$

Since $(q; q)_k \geq (q; q)_\infty > 0$, the tail of (6.1) is bounded by the right side of (6.3). Hence the series converges locally uniformly to an analytic function g . For each fixed node cq^j , every partial sum with $n \geq j$ equals $f(cq^j)$; therefore $g(cq^j) = f(cq^j)$ for all j . The difference $g - f$ is analytic and has zeros accumulating at 0, so it vanishes identically. The same tail estimate gives the displayed error bound. $\square$

Remark 6.2 (A convergence theorem, not a stability theorem). The bound is local in a disk $|x| < R$ on which f is analytic. It does not contradict the global Lebesgue growth proved later. Analytic data generated from one exact function can undergo cancellations that arbitrary perturbed sample vectors do not. Moreover, the Fabius function is flat and nonanalytic at 0, so [Theorem 6.1](#) deliberately does not cover the most interesting dyadic specialization; that case needs its own arithmetic estimates.

6.2 Complex bases and spiralling combs

The finite algebra survives for complex q whenever the nodes are distinct. For $|q| < 1$, the same proof gives a complex convergence theorem after replacing every occurrence of q in a majorant by $|q|$ and assuming that the disk containing the comb lies inside the analytic domain. The nodes then approach 0 along a logarithmic spiral. Lebesgue constants cease to be orderable real quantities on an interval, but the norm of the evaluation functional at a chosen point remains meaningful. The spiral case is a natural setting for analytic continuation from exponentially separated samples and is listed as a research direction in [Chapter 13](#).

6.3 Meromorphic targets

If f is meromorphic near the closed disk and its poles are known, one may subtract their principal parts before applying the Newton series. More systematically, a rational Newton–Padé construction can preserve the geometric sampling while moving the approximation space from polynomials to rational functions. The global stability theorem below strongly suggests that such rationalization is not optional when accurate reconstruction is required across the persistent gap.

A sharp stability dichotomy

7.1 Lebesgue functions

For real nodes $c, cq, \dots, cq^n$, define

$$\Lambda_n(x) := \sum_{j=0}^n |\ell_{j,n}(x)|, \quad \Lambda_n^* := \max_{0 \leq x \leq c} \Lambda_n(x). \quad (7.1)$$

The operator norm of interpolation from nodal ℓ^∞ data to the value at x is exactly $\Lambda_n(x)$; the global norm on $C[0, c]$ is Λ_n^* . By affine covariance, both quantities are independent of c after x is scaled by c .

7.2 Bounded extrapolation to the accumulation point

Theorem 7.1 (Exact accumulation-point Lebesgue constant). *For $0 < q < 1$,*

$$\Lambda_n(0) = \frac{(-q; q)_n}{(q; q)_n}. \quad (7.2)$$

In particular,

$$\Lambda_n(0) \longrightarrow \Lambda_\infty(0) := \frac{(-q; q)_\infty}{(q; q)_\infty} < \infty. \quad (7.3)$$

Proof. By (4.7), with $r = n - j$,

$$\begin{aligned} \Lambda_n(0) &= \sum_{j=0}^n \frac{q^{\binom{n-j+1}{2}}}{(q; q)_j (q; q)_{n-j}} \\ &= \frac{1}{(q; q)_n} \sum_{r=0}^n q^{\binom{r+1}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q. \end{aligned}$$

The finite q -binomial theorem with parameter q gives

$$\sum_{r=0}^n q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q q^r = (-q; q)_n.$$

This proves the finite formula. Both infinite products converge to positive limits for $0 < q < 1$. $\square$

Thus geometric Richardson extrapolation to the point where the grid accumulates has a uniformly bounded worst-case amplification. For the half-base,

$$\Lambda_\infty(0) = \frac{(-1/2; 1/2)_\infty}{(1/2; 1/2)_\infty} = 8.2559879357782500655 \dots \quad (7.4)$$

The number is not tiny, but it is independent of degree.

7.3 Boundary-layer profile near the isolated outer node

Set $c = 1$ and write $M_n = \binom{n}{2}$. The cardinal belonging to the smallest node q^n has, in the top gap $q < x < 1$, the exact form

$$|\ell_{n,n}(x)| = q^{-M_n} (1-x) x^{n-1} \frac{(q/x; q)_{n-1}}{(q; q)_n}. \quad (7.5)$$

Although this cardinal is attached to the node closest to zero, it becomes enormous near the opposite endpoint. The other small-node cardinals form a universal multiplicative cloud around it.

Theorem 7.2 (Scaled Lebesgue boundary profile). *Fix $0 < q < 1$. Uniformly for α in compact subsets of $(0, \infty)$,*

$$\boxed{nq^{M_n} \Lambda_n(1 - \alpha/n) \longrightarrow 2(-q; q)_\infty \alpha e^{-\alpha}}. \quad (7.6)$$

Proof. First, (7.5) gives

$$nq^{M_n} |\ell_{n,n}(1 - \alpha/n)| = \alpha (1 - \alpha/n)^{n-1} \frac{(q/(1 - \alpha/n); q)_{n-1}}{(q; q)_n}. \quad (7.7)$$

The first two factors tend uniformly to $\alpha e^{-\alpha}$. For the product ratio, take logarithms. On a compact α -set and for large n , the mean-value theorem gives

$$|\log(1 - q^r/(1 - \alpha/n)) - \log(1 - q^r)| \leq \frac{Cq^r}{n}.$$

after increasing C to handle the finitely many small r . Summation over r is $O(n^{-1})$, and the extra factor $(1 - q^n)^{-1}$ tends to one. Hence the product ratio in (7.7) tends uniformly to one.

Now compare every cardinal to the last one. Put $j = n - r$. The barycentric weights in (4.2) imply

$$\frac{|\ell_{n-r,n}(x)|}{|\ell_{n,n}(x)|} = q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q \frac{|x - q^n|}{|x - q^{n-r}|}. \quad (7.8)$$

For $x = 1 - \alpha/n$ and fixed r , the right side tends to $q^{\binom{r}{2}}/(q; q)_r$. For $0 \leq r \leq n - 1$, the denominator $x - q^{n-r}$ is uniformly bounded below because $q^{n-r} \leq q$, and

$$\begin{bmatrix} n \\ r \end{bmatrix}_q \leq \frac{1}{(q; q)_r} \leq \frac{1}{(q; q)_\infty}.$$

The summable majorant $Cq^{\binom{r}{2}}$ permits dominated convergence. The remaining term $r = n$, corresponding to the outer node 1, is $O(nq^{M_n})$ relative to the last cardinal and therefore vanishes. Hence

$$\begin{aligned} \frac{\Lambda_n(1 - \alpha/n)}{|\ell_{n,n}(1 - \alpha/n)|} &\longrightarrow \sum_{r=0}^{\infty} \frac{q^{\binom{r}{2}}}{(q; q)_r} \\ &= (-1; q)_\infty = 2(-q; q)_\infty, \end{aligned}$$

by Euler's identity. Combining this with (7.7) proves the theorem. $\square$

7.4 The global asymptotic

Theorem 7.3 (Sharp global Lebesgue constant). *Let $0 < q < 1$, and let x_n^* be any maximizer of Λ_n on $[0, c]$. Then*

$$\Lambda_n^* \sim \frac{2(-q; q)_\infty}{e} \frac{q^{-\binom{n}{2}}}{n}, \quad (7.9)$$

$$\frac{x_n^*}{c} = 1 - \frac{1}{n} + o(n^{-1}). \quad (7.10)$$

Proof. By affine covariance take $c = 1$. The lower bound follows from [Theorem 7.2](#) at $\alpha = 1$.

We first show that every maximizing sequence approaches 1. Fix $\varepsilon > 0$. With $j = n - r$, the explicit cardinal formula gives

$$q^{M_n} |\ell_{n-r, n}(x)| = q^{\binom{r}{2}} \frac{\prod_{s \neq n-r} |x - q^s|}{(q; q)_{n-r} (q; q)_r}. \quad (7.11)$$

For $0 \leq x \leq 1 - \varepsilon$, choose R with $q^R \leq \varepsilon/2$. All but at most one of the factors with $s \geq R$ are at most $1 - \varepsilon/2$. Since each denominator is at least $(q; q)_\infty$, summing (7.11) over r gives

$$q^{M_n} \Lambda_n(x) \leq C_{q, \varepsilon} (1 - \varepsilon/2)^{n-R-1}.$$

After multiplication by n , this tends to zero, whereas the lower bound at $1 - 1/n$ tends to the positive constant $2(-q; q)_\infty/e$. Thus $x_n^* \rightarrow 1$.

Put $\delta_n = 1 - x_n^*$. On $x \geq x_0 := (1 + q)/2$, the outer-node cardinal satisfies

$$|\ell_{0, n}(x)| = \prod_{s=1}^n \frac{x - q^s}{1 - q^s} \leq 1.$$

For the remaining cardinals, (7.8) and the summability of $q^{\binom{r}{2}}$ give

$$\Lambda_n(x) \leq 1 + C_q |\ell_{n, n}(x)|.$$

Moreover, every factor of $(q/x; q)_{n-1}$ is no larger than the corresponding factor of $(q; q)_{n-1}$, so

$$|\ell_{n, n}(x)| \leq \frac{q^{-M_n}}{1 - q} (1 - x)x^{n-1}. \quad (7.12)$$

It follows that

$$nq^{M_n} \Lambda_n(x_n^*) \leq nq^{M_n} + C'_q(n\delta_n)(1 - \delta_n)^{n-1}.$$

If $n\delta_n \rightarrow 0$ or $n\delta_n \rightarrow \infty$, the right side tends to zero, contradicting the positive lower bound. Hence every subsequence has a further subsequence with $n\delta_n \rightarrow \alpha \in (0, \infty)$. The locally uniform profile theorem then gives

$$\lim nq^{M_n} \Lambda_n(x_n^*) = 2(-q; q)_\infty \alpha e^{-\alpha} \leq \frac{2(-q; q)_\infty}{e}.$$

The lower bound at $\alpha = 1$ forces equality. Since $\alpha e^{-\alpha}$ has a unique maximum at $\alpha = 1$, every subsequential limit is 1. This proves both assertions. $\square$

For $q = 1/2$ the theorem becomes

$$\Lambda_n^* \sim 1.7542191571673478084 \dots \frac{2^{\binom{n}{2}}}{n}. \quad (7.13)$$

The coexistence of (7.4) and (7.13) is the central stability fact of this report.

7.5 Interpretation: stable at the accumulation end, unstable in the gap

The geometry has two radically different regimes.

1. Near 0, every new node is closer to the evaluation point, and the alternating Richardson weights have bounded total variation.
2. Near c , only one node lies at the outer endpoint while the remaining nodes retreat geometrically toward zero. The last few small-node cardinals form the Euler cloud $2(-q; q)_\infty$ and reach their maximum in a layer of width c/n .

The instability is stronger than ordinary exponential growth: its logarithm is quadratic in n . No choice between Newton, Lagrange, and barycentric representations can change this operator norm. Those representations can only determine whether a numerical algorithm adds avoidable roundoff error to the unavoidable sensitivity of the interpolation problem.

Jackson integration and confluent geometric data

The same basis that triangularizes sampling on $cq^{\mathbb{N}_0}$ also triangularizes Jackson integration. This gives a useful dual picture: geometric Newton coefficients are simultaneously interpolation coordinates and exact q -quadrature coordinates.

8.1 The Newton–Jackson integral formula

Recall the Jackson integral on $[0, c]$,

$$\int_0^c f(x) \, d_q x := (1 - q)c \sum_{j=0}^{\infty} q^j f(cq^j), \quad 0 < q < 1. \quad (8.1)$$

The definition makes the geometric comb, including its multiplicative weights, into the native quadrature grid of q -calculus [9, 5].

Theorem 8.1 (Jackson moments of the geometric Newton basis). *For every $k \geq 0$,*

$$\int_0^c N_k(x; c, q) \, d_q x = (-1)^k c^{k+1} \frac{q^{\binom{k+1}{2}}}{[k+1]_q}. \quad (8.2)$$

Consequently, if $p = \sum_{k=0}^d \gamma_k N_k$ is a polynomial, then

$$\int_0^c p(x) \, d_q x = \sum_{k=0}^d \gamma_k (-1)^k c^{k+1} \frac{q^{\binom{k+1}{2}}}{[k+1]_q}. \quad (8.3)$$

Proof. At a comb point $x = cq^j$ the factor $N_k(cq^j)$ vanishes for $j < k$. For $j \geq k$,

$$N_k(cq^j) = (-1)^k c^k q^{\binom{k}{2}} \frac{(q; q)_j}{(q; q)_{j-k}}. \quad (8.4)$$

Therefore

$$\begin{aligned} \int_0^c N_k(x) \, d_q x &= (-1)^k c^{k+1} q^{\binom{k}{2}} (1 - q) \sum_{j=k}^{\infty} q^j \frac{(q; q)_j}{(q; q)_{j-k}} \\ &= (-1)^k c^{k+1} q^{\binom{k}{2}+k} (1 - q) (q; q)_k \sum_{r=0}^{\infty} q^r \frac{(q^{k+1}; q)_r}{(q; q)_r}. \end{aligned}$$

The infinite q -binomial theorem gives

$$\sum_{r \geq 0} q^r \frac{(q^{k+1}; q)_r}{(q; q)_r} = \frac{(q^{k+2}; q)_\infty}{(q; q)_\infty} = \frac{1}{(q; q)_{k+1}}.$$

Since $(1 - q)/(1 - q^{k+1}) = 1/[k + 1]_q$, this proves (8.2). Linearity proves (8.3). $\square$

Corollary 8.2 (An analytic Newton–Jackson series). *Under the hypotheses of Theorem 6.1, if $f(x) = \sum_{k \geq 0} \gamma_k N_k(x; c, q)$ on a neighborhood of $[0, c]$, then*

$$\int_0^c f(x) \, d_q x = \sum_{k=0}^{\infty} \gamma_k (-1)^k c^{k+1} \frac{q^{\binom{k+1}{2}}}{[k + 1]_q}, \quad (8.5)$$

and the series converges absolutely.

Proof. Uniform convergence on $[0, c]$ permits termwise Jackson integration; the absolute estimate in Theorem 6.1, together with the extra factor $q^{\binom{k+1}{2}}$, gives absolute convergence directly. $\square$

The quadratic exponent in (8.5) is important. Even when the interpolation coefficients grow moderately, Jackson integration strongly damps the high Newton levels. Interpolation and quadrature therefore have very different condition profiles on the same comb.

8.2 Repeated geometric nodes

Ordinary Hermite interpolation replaces some simple nodes by repeated nodes. If every point of the finite comb has multiplicity s , the nodal polynomial remains a single q -product:

$$\prod_{j=0}^n (x - cq^j)^s = \left((-c)^{n+1} q^{\binom{n+1}{2}} (x/c; q^{-1})_{n+1} \right)^s. \quad (8.6)$$

Thus the q -Pochhammer structure survives confluence at the level of the annihilating polynomial, although the individual cardinal functions acquire higher-order principal parts.

For multiplicities $s_j \geq 1$, put

$$\Omega(x) = \prod_{j=0}^n (x - cq^j)^{s_j}.$$

The Hermite basis attached to the derivative functional $p \mapsto p^{(r)}(cq^j)$, $0 \leq r < s_j$, can be obtained by taking the first s_j terms in the Taylor expansion at cq^j of

$$\frac{(x - cq^j)^{s_j}}{\Omega(x)}. \quad (8.7)$$

All denominators in these coefficients are products of powers of $q^j - q^r$ and hence reduce to q -Pochhammer symbols. This gives a systematic, division-explicit route to geometric Hermite formulae without forming a confluent Vandermonde matrix.

Remark 8.3. A fully expanded formula is naturally indexed by compositions of the derivative order among the factors $(x - cq^r)^{-s_r}$. Complete homogeneous symmetric functions again appear, now on a multiset in which $(cq^j - cq^r)^{-1}$ occurs s_r times. This is the confluent analogue of the complete-homogeneous residual calculus in Chapter 5.

8.3 Anchoring ordinary derivatives at the accumulation point

For the Fabius and Rvachev applications it is often more useful to prescribe ordinary derivatives at the accumulation point 0 than to repeat every comb node. The following factorization converts that mixed Hermite problem into one ordinary geometric interpolation problem.

Theorem 8.4 (Accumulation-point Hermite factorization). *Let f have derivatives through order m at 0, and define*

$$T_m f(x) = \sum_{r=0}^m \frac{f^{(r)}(0)}{r!} x^r, \quad g_m(x) = \frac{f(x) - T_m f(x)}{x^{m+1}}, \quad (8.8)$$

where $g_m(0)$ is interpreted by continuous extension when it exists. Let $\mathcal{I}_n^{c,q}$ denote interpolation at $c, cq, \dots, cq^n$. Then

$$\boxed{H_{n,m}^{c,q} f(x) := T_m f(x) + x^{m+1} \mathcal{I}_n^{c,q} g_m(x)} \quad (8.9)$$

is the unique polynomial of degree at most $n + m + 1$ satisfying

$$(H_{n,m}^{c,q} f)^{(r)}(0) = f^{(r)}(0), \quad 0 \leq r \leq m, \quad (8.10)$$

$$H_{n,m}^{c,q} f(cq^j) = f(cq^j), \quad 0 \leq j \leq n. \quad (8.11)$$

Its exact error is

$$f(x) - H_{n,m}^{c,q} f(x) = x^{m+1} (g_m(x) - \mathcal{I}_n^{c,q} g_m(x)). \quad (8.12)$$

Proof. The difference between (8.9) and $T_m f$ is divisible by x^{m+1} , proving all jet conditions. At a nonzero comb node, interpolation of g_m gives

$$T_m f(cq^j) + (cq^j)^{m+1} g_m(cq^j) = f(cq^j).$$

If two polynomials of degree at most $n + m + 1$ satisfy the same conditions, their difference is divisible by x^{m+1} and by $N_{n+1}(x; c, q)$. These relatively prime factors have total degree $n + m + 2$, so the difference is zero. Subtracting (8.9) from f gives (8.12). $\square$

Corollary 8.5 (Analytic convergence with prescribed jets). *If f is analytic in a disk containing $[0, c]$, then for every fixed m , $H_{n,m}^{c,q} f \rightarrow f$ locally uniformly in that disk as $n \rightarrow \infty$. Moreover, all derivatives through order m agree exactly for every n .*

The factorization also applies to nonanalytic but flat functions. If $f^{(r)}(0) = 0$ for every r , then

$$H_{n,m}^{c,q} f(x) = x^{m+1} \mathcal{I}_n^{c,q} \left(\frac{f(x)}{x^{m+1}} \right). \quad (8.13)$$

It is tempting to let m grow with n . The dyadic-comb experiments in the neighboring equispaced setting show, however, that too many exact derivative constraints can reintroduce a different large-cardinal mode. For a geometric comb, the correct joint scaling of m and n is an open conditioning problem; see [Theorem 8.6](#).

Research problem 8.6 (Optimal geometric Hermite anchoring). For fixed $0 < q < 1$ and a target with a flat germ at 0, determine the largest jet order $m = m(n)$ for which the operators $H_{n,m}^{c,q}$ remain stable on a specified compact subset, and determine whether a transition occurs near $m \asymp n$, $m \asymp n / \log n$, or another scale.

Part III

The dyadic Fabius–Rvachev specialization

Fabius moments and the exact half-base Gaussian transform

Set $q = 1/2$ throughout this chapter. The geometric comb $1, q, q^2, \dots$ is then the inverse-dyadic scale selected by the defining self-similarity of the Fabius function. The interpolation data are not arbitrary: after a quadratic normalization, they are the moment coefficients of a compactly supported probability distribution. This produces a second, independent triangular structure on top of the Gaussian Pascal structure.

9.1 Probability model, moments, and the preferred normalization

Let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[0, 1]$, and put

$$X = \sum_{r=1}^{\infty} 2^{-r} U_r. \quad (9.1)$$

The random series takes values in $[0, 1]$. Its distribution function is the Fabius function F ; this probability model is equivalent to the customary functional-differential definition [6, 2]. The symmetry $X \stackrel{d}{=} 1 - X$ gives

$$F(x) + F(1 - x) = 1, \quad 0 \leq x \leq 1. \quad (9.2)$$

On $[-1, 1]$ the Rvachev function is related by

$$\text{up}(t) = F(1 - |t|). \quad (9.3)$$

Define the moments and their exponential normalization by

$$d_n := \mathbb{E}X^n, \quad a_n := \frac{d_n}{n!}. \quad (9.4)$$

The notation d_n is especially natural for the Rvachev half-moments, because

$$d_n = n \int_0^1 t^{n-1} \text{up}(t) \, dt. \quad (9.5)$$

Indeed, the layer-cake identity and (9.2) give

$$\mathbb{E}X^n = n \int_0^1 t^{n-1} \mathbb{P}(X > t) \, dt = n \int_0^1 t^{n-1} F(1 - t) \, dt.$$

Thus d_n is the preferred moment notation throughout the dyadic specialization.

The moment-generating function is the entire product

$$A(z) := \mathbb{E}e^{zX} = \sum_{n=0}^{\infty} a_n z^n \quad (9.6)$$

$$= \prod_{r=1}^{\infty} \frac{e^{z/2^r} - 1}{z/2^r}. \quad (9.7)$$

Separating the first factor gives

$$A(2z) = \frac{e^z - 1}{z} A(z). \quad (9.8)$$

Proposition 9.1 (Exact moment recurrence). *The coefficients $a_n \in \mathbb{Q}$ are determined by $a_0 = 1$ and*

$$(2^n - 1)a_n = \sum_{j=0}^{n-1} \frac{a_j}{(n-j+1)!}, \quad n \geq 1. \quad (9.9)$$

In particular $0 < a_n \leq 1/n!$.

Proof. Compare coefficients of z^n in (9.8):

$$2^n a_n = \sum_{j=0}^n \frac{a_j}{(n-j+1)!}.$$

Moving the $j = n$ term proves the recurrence. Rationality follows by induction. Since $0 \leq X \leq 1$, one has $0 < d_n \leq 1$, hence the stated bound. $\square$

9.2 Inverse-dyadic values are normalized moments

The Fabius differential equation is

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}. \quad (9.10)$$

Repeated differentiation on the shrinking interval gives

$$F^{(n)}(x) = 2^{\binom{n+1}{2}} F(2^n x), \quad 0 \leq x \leq 2^{-n}. \quad (9.11)$$

All derivatives of F vanish at 0.

Theorem 9.2 (Moment formula for inverse-dyadic values). *For every $n \geq 0$,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} a_n = 2^{-\binom{n}{2}} \frac{d_n}{n!}. \quad (9.12)$$

Proof. The cases $n = 0$ and $n = 1$ are immediate. For $n \geq 1$, Taylor's formula with integral remainder and the flat germ at zero give

$$\begin{aligned} F(2^{-n}) &= \frac{1}{(n-1)!} \int_0^{2^{-n}} (2^{-n} - t)^{n-1} F^{(n)}(t) \, dt \\ &= \frac{2^{-\binom{n}{2}}}{(n-1)!} \int_0^1 (1-u)^{n-1} F(u) \, du. \end{aligned}$$

After $t = 1 - u$, the last integral is d_n/n by (9.5). The result follows. $\square$

The first values are

$$F(1) = 1, \quad F(1/2) = \frac{1}{2}, \quad F(1/4) = \frac{5}{72}, \quad F(1/8) = \frac{1}{288}, \quad F(1/16) = \frac{143}{2073600}. \quad (9.13)$$

Their rapidly growing denominators are partly neutralized by the quadratic factor $2^{\binom{n}{2}}$, leaving the much more regular moment sequence a_n .

9.3 The exact Gaussian transform of Fabius data

Let ν_k be the coefficient of $N_k(x; 1, q)$ in the Newton interpolant to F at $1, q, \dots, q^k$:

$$\nu_k = [F; 1, q, \dots, q^k]. \quad (9.14)$$

Theorem 9.3 (Half-base Fabius–Gaussian transform). *For $q = 1/2$,*

$$\nu_k = \frac{1}{(q; q)_k} \sum_{j=0}^k (-1)^j \begin{bmatrix} k \\ j \end{bmatrix}_{q^{-1}} a_j = \frac{1}{(1/2; 1/2)_k} \sum_{j=0}^k (-1)^j \begin{bmatrix} k \\ j \end{bmatrix}_2 a_j. \quad (9.15)$$

Consequently the degree- n interpolant is

$$(\mathcal{I}_n F)(x) = \sum_{k=0}^n \nu_k \prod_{r=0}^{k-1} (x - 2^{-r}). \quad (9.16)$$

Every coefficient is rational.

Proof. Insert $F(q^j) = q^{\binom{j}{2}} a_j$ from [Theorem 9.2](#) into the explicit divided difference (2.7). The exponent simplifies:

$$-jk + \binom{j+1}{2} + \binom{j}{2} = j(j-k).$$

Reciprocity of Gaussian coefficients gives

$$\begin{bmatrix} k \\ j \end{bmatrix}_{q^{-1}} = q^{-j(k-j)} \begin{bmatrix} k \\ j \end{bmatrix}_q = q^{j(j-k)} \begin{bmatrix} k \\ j \end{bmatrix}_q.$$

Factoring $(q; q)_k$ proves (9.15). The Newton formula and rationality are immediate. $\square$

This transform is a meeting point of three triangular arrays:

1. the moment recurrence (9.9);
2. the reciprocal-base Gaussian Pascal array $\begin{bmatrix} k \\ j \end{bmatrix}_2$;
3. the Newton evaluation array on $1, 1/2, 1/4, \dots$

The resulting coefficients need not be small. Exact values begin

$$\nu_0 = 1, \quad \nu_1 = 1, \quad \nu_2 = -\frac{26}{27}, \quad \nu_3 = -\frac{128}{27}, \quad (9.17)$$

$$\nu_4 = -\frac{4253248}{637875}, \quad \nu_5 = \frac{189673472}{19774125}. \quad (9.18)$$

The signs and magnitudes already foreshadow the gap instability studied in [Chapter 10](#).

9.4 A finite Gaussian-cardinal formula

The Lagrange form packages the same transform into one rational kernel.

Theorem 9.4 (Finite Gaussian formula for the Fabius interpolant). *Let*

$$\omega_{n+1}(x) = \prod_{r=0}^n (x - q^r), \quad q = \frac{1}{2}.$$

Then

$$(\mathcal{I}_n F)(x) = \frac{\omega_{n+1}(x)}{(q; q)_n} \sum_{j=0}^n \frac{(-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}} a_j}{x - q^j}. \quad (9.19)$$

At a node the right side is interpreted by cancellation. Formula (9.19) is an identity in $\mathbb{Q}[x]$.

Proof. The Lagrange weight at q^j is

$$\lambda_{j,n} = \frac{(-1)^j q^{-nj + \binom{j+1}{2}}}{(q; q)_j (q; q)_{n-j}}.$$

Multiplication by $F(q^j) = q^{\binom{j}{2}} a_j$ changes the exponent to $j(j-n)$. Since

$$\frac{q^{j(j-n)}}{(q; q)_j (q; q)_{n-j}} = \frac{1}{(q; q)_n} \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}},$$

the ordinary Lagrange formula gives the result. $\square$

The numerator in (9.19) is a finite base-2 Gaussian transform of the regularized moments. The poles are only an encoding device: the zero of ω_{n+1} cancels each one exactly.

Boundary recovery and gap divergence

The two stability regimes of Chapter 7 become unusually vivid for the Fabius data. At the accumulation endpoint, interpolation recovers not only $F(0) = 0$ but every fixed zero derivative. In the persistent gap $(1/2, 1)$, the same exact rational polynomials grow at a nearly $\exp((\log 2)n^2/4)$ scale.

10.1 The boundary residue and its generating function

Define

$$R_n := (\mathcal{I}_n F)(0). \quad (10.1)$$

Using the endpoint cardinal weights from Theorem 4.3 gives the exact finite sum below.

Theorem 10.1 (Fabius boundary residue). *For $q = 1/2$,*

$$R_n = \sum_{j=0}^n \frac{(-1)^{n-j} q^{\binom{n-j+1}{2} + \binom{j}{2}}}{(q; q)_j (q; q)_{n-j}} a_j. \quad (10.2)$$

Its ordinary generating function is

$$\sum_{n=0}^{\infty} R_n z^n = (qz; q)_{\infty} \sum_{j=0}^{\infty} \frac{q^{\binom{j}{2}} a_j}{(q; q)_j} z^j. \quad (10.3)$$

Both factors on the right are entire functions of z .

Proof. The endpoint cardinal is

$$\ell_{j,n}(0) = \frac{(-1)^{n-j} q^{\binom{n-j+1}{2}}}{(q; q)_j (q; q)_{n-j}}.$$

Multiplication by $F(q^j) = q^{\binom{j}{2}} a_j$ proves (10.2). Euler's expansion

$$(qz; q)_{\infty} = \sum_{r=0}^{\infty} \frac{(-1)^r q^{\binom{r+1}{2}}}{(q; q)_r} z^r$$

shows that the coefficient of z^n in the product on the right of (10.3) is precisely (10.2). The first factor is a normally convergent infinite product. In the second, $a_j \leq 1/j!$ and the factor $q^{\binom{j}{2}}$ gives more than enough decay for entire convergence. $\square$

Equation (10.3) may be viewed as a q -Borel regularization of the Fabius moment series, followed by multiplication by the Euler zero product. It explains why the boundary residues decay far faster than any geometric sequence.

Theorem 10.2 (Superexponential boundary recovery). *For every $n \geq 0$,*

$$|R_n| \leq \frac{e}{(q; q)_\infty^2} q^{\lfloor n^2/4 \rfloor}, \quad q = \frac{1}{2}. \quad (10.4)$$

In particular $R_n \rightarrow F(0) = 0$ superexponentially.

Proof. Since $a_j \leq 1/j!$ and each finite $(q; q)_r$ is at least $(q; q)_\infty$,

$$|R_n| \leq \frac{1}{(q; q)_\infty^2} \sum_{j=0}^n \frac{q^{E_n(j)}}{j!}, \quad E_n(j) = \binom{n-j+1}{2} + \binom{j}{2}.$$

A direct completion of the square gives $E_n(j) \geq \lfloor n^2/4 \rfloor$. Finally $\sum_{j=0}^n 1/j! \leq e$. $\square$

10.2 Recovery of every fixed flat jet

The value estimate extends to arbitrary fixed derivative order. For a finite set Y , write $e_m(Y)$ for its elementary symmetric polynomial of degree m .

Theorem 10.3 (Exact derivative formula and fixed-jet bound). *Let $0 \leq m \leq n$. Then*

$$(\mathcal{I}_n F)^{(m)}(0) = (-1)^m m! \sum_{j=0}^n \ell_{j,n}(0) q^{\binom{j}{2}} a_j e_m(\{q^{-r} : 0 \leq r \leq n, r \neq j\}). \quad (10.5)$$

Moreover,

$$\left| (\mathcal{I}_n F)^{(m)}(0) \right| \leq \frac{em!}{(q; q)_\infty^2} \binom{n}{m} q^{\lfloor n^2/4 \rfloor - mn + \binom{m}{2}}. \quad (10.6)$$

For every fixed m ,

$$(\mathcal{I}_n F)^{(m)}(0) \longrightarrow F^{(m)}(0) = 0. \quad (10.7)$$

Proof. Factor a cardinal at zero as

$$\ell_{j,n}(x) = \ell_{j,n}(0) \prod_{\substack{0 \leq r \leq n \\ r \neq j}} (1 - xq^{-r}).$$

The coefficient of x^m is $(-1)^m \ell_{j,n}(0) e_m(\{q^{-r} : r \neq j\})$, proving (10.5) after summing against the nodal values.

There are $\binom{n}{m}$ products in the elementary symmetric polynomial, and each is bounded by the product of the m largest members of $\{q^{-r} : 0 \leq r \leq n\}$:

$$e_m(\{q^{-r} : r \neq j\}) \leq \binom{n}{m} q^{-mn + \binom{m}{2}}. \quad (10.8)$$

Apply the proof of Theorem 10.2. For fixed m , the quadratic term $\lfloor n^2/4 \rfloor$ dominates the linear loss mn and the polynomial factor $\binom{n}{m}$, proving convergence. $\square$

This result is stronger than value convergence but weaker than convergence of all jets simultaneously. The order m must remain fixed while $n \rightarrow \infty$. If m grows proportionally to n , the exponent in (10.6) changes sign, and a new asymptotic problem begins.

Corollary 10.4 (Boundary jets for the Rvachev function). *Let*

$$J_n(t) := (\mathcal{I}_n F)(1-t). \quad (10.9)$$

Then $J_n(1 - 2^{-j}) = \text{up}(1 - 2^{-j})$ for $0 \leq j \leq n$, and for every fixed m ,

$$J_n^{(m)}(1) \longrightarrow \text{up}^{(m)}(1) = 0. \quad (10.10)$$

10.3 A rigorous gap-growth upper bound

Fix $x \in (q, 1) = (1/2, 1)$. This point remains separated from all nodes except the isolated node 1. The finite Gaussian formula gives a substantially sharper data-specific estimate than the full Lebesgue constant.

Theorem 10.5 (Quadratic-exponential gap bound). *For fixed $x \in (q, 1)$,*

$$|(\mathcal{I}_n F)(x)| \leq C_{q,x} x^n q^{-\lfloor n^2/4 \rfloor}, \quad (10.11)$$

where one may take

$$C_{q,x} = \frac{1}{(q; q)_\infty^2} \left(1 + \frac{e(1-x)}{x-q} \right). \quad (10.12)$$

A sharper logarithmic estimate is

$$\log^+ |(\mathcal{I}_n F)(x)| \leq \frac{\log(1/q)}{4} n^2 - \frac{n}{2} \log n + O_{q,x}(n + (\log n)^2). \quad (10.13)$$

Proof. For $x \in (q, 1)$,

$$|\omega_{n+1}(x)| = (1-x) \prod_{r=1}^n (x - q^r) \leq (1-x)x^n.$$

In (9.19), the $j = 0$ denominator is $1 - x$, whereas $x - q^j \geq x - q$ for $j \geq 1$. Also

$$\frac{1}{(q; q)_n} \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}} = \frac{q^{-j(n-j)}}{(q; q)_j (q; q)_{n-j}}, \quad a_j \leq \frac{1}{j!}.$$

Taking absolute values and using $j(n-j) \leq \lfloor n^2/4 \rfloor$ proves (10.11).

For the sharper form, it suffices to estimate

$$S_n = \sum_{j=0}^n \frac{q^{-j(n-j)}}{j!}.$$

With $L = \log(1/q)$ and $t_j = \exp(Lj(n-j))/j!$,

$$\frac{t_{j+1}}{t_j} = \frac{\exp(L(n-2j-1))}{j+1}. \quad (10.14)$$

This ratio is strictly decreasing, so the sequence is unimodal. Its maximizer j_n satisfies

$$j_n = \frac{n}{2} - \frac{\log n}{2L} + O_q(1). \quad (10.15)$$

Stirling's formula at this index gives

$$\log t_{j_n} = \frac{L}{4} n^2 - \frac{n}{2} \log n + O_q(n + (\log n)^2).$$

Since $S_n \leq (n+1)t_{j_n}$, substitution into the preceding cardinal bound proves (10.13). $\square$

The dominant data index is not exactly $n/2$: the factorial normalization of the Fabius moments shifts it left by approximately $(2 \log 2)^{-1} \log n$. This is a genuine competition between the reciprocal Gaussian coefficient $2^{j(n-j)}$ and the moment decay $1/j!$.

10.4 A saddle-point conjecture for the actual interpolation error

The theorem above discards alternating signs. Exact rational experiments show that cancellation changes the lower-order terms but apparently not the two leading scales.

Conjecture 10.6 (Fabius gap asymptotic). For every fixed $x \in (1/2, 1)$ outside a possibly discrete exceptional set,

$$\log |(\mathcal{I}_n F)(x) - F(x)| = \frac{\log 2}{4} n^2 - \frac{n}{2} \log n + O_x(n). \quad (10.16)$$

More precisely, the $O_x(n)$ term should contain an explicit linear coefficient and a bounded discrete-saddle modulation determined by the fractional part of

$$\frac{n}{2} - \frac{\log n}{2 \log 2} + \sigma(x). \quad (10.17)$$

A proof will require a signed saddle analysis of the finite sum (9.19), with sufficiently uniform asymptotics for a_j . The moment product (9.7) suggests that Hayman-type coefficient methods or a two-dimensional saddle combining the moment coefficient and Gaussian kernel may be effective.

Remark 10.7 (Why endpoint flatness does not prevent the gap catastrophe). The interpolation nodes reveal the flat germ with exceptional accuracy: [Theorem 10.3](#) recovers every fixed derivative at zero. Yet those same nodes leave the interval $(1/2, 1)$ permanently unsampled. A polynomial forced through increasingly many points near zero must transport their tiny but arithmetically structured values across this fixed gap. The reciprocal Gaussian coefficients provide precisely the required, and highly unstable, transport.

The Rvachev synthesis loop in geometric coordinates

Interpolation produces polynomials; Rvachev theory reproduces polynomials by compactly supported atoms. Composing the two operations gives an exact finite factorization of geometric-comb interpolation through translates of up . Because the geometric cardinal coefficients are Gaussian and the polynomial reproduction coefficients are inverse-moment Appell values, the result is naturally a Gaussian–Appell transform.

11.1 Fourier zeros and inverse-moment Appell polynomials

Use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx.$$

For $u = \text{up}$,

$$\widehat{u}(\xi) = \prod_{r=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^r}\right), \quad \text{sinc}(z) = \frac{\sin z}{z}. \quad (11.1)$$

Hence, for every nonzero integer m ,

$$\text{ord}_{\xi=m} \widehat{u}(\xi) = 1 + \nu_2(m). \quad (11.2)$$

This exact multiplicity is the dyadic Strang–Fix mechanism behind the reproduction theorem below [11, 12].

The moment-generating function of the centered random series $Z = \sum_{r \geq 1} 2^{-r} V_r$, with $V_r \sim \text{Unif}[-1, 1]$, is

$$\mathcal{M}_u(z) = \int_{-1}^1 e^{zx} u(x) dx = \prod_{r=1}^{\infty} \frac{\sinh(z/2^r)}{z/2^r}. \quad (11.3)$$

Its cumulants are

$$\log \mathcal{M}_u(z) = \sum_{r=1}^{\infty} \kappa_{2r} \frac{z^{2r}}{(2r)!}, \quad \kappa_{2r} = \frac{B_{2r}}{2r(1 - 4^{-r})}. \quad (11.4)$$

Define rational inverse moments γ_r by

$$\frac{1}{\mathcal{M}_u(z)} = \sum_{r=0}^{\infty} \gamma_r \frac{z^r}{r!}. \quad (11.5)$$

Thus $\gamma_{2r+1} = 0$, and

$$\gamma_0 = 1, \quad \gamma_2 = -\frac{1}{9}, \quad \gamma_4 = \frac{31}{675}, \quad \gamma_6 = -\frac{2347}{59535}. \quad (11.6)$$

For a polynomial p , introduce the terminating operator

$$\mathcal{A}p := \mathcal{M}_u(D)^{-1}p = \sum_{r=0}^{\lfloor \deg p/2 \rfloor} \frac{\gamma_{2r}}{(2r)!} p^{(2r)}. \quad (11.7)$$

It is triangular with unit diagonal and therefore an automorphism on every finite polynomial space.

11.2 Exact fixed-radius polynomial reproduction

Theorem 11.1 (Dyadic fixed-radius synthesis). *Let p have degree at most n , and set*

$$m = 2^n, \quad h = 2^{-n}, \quad K_n = \{-2m + 1, -2m + 2, \dots, 2m - 1\}. \quad (11.8)$$

Then

$$p(x) = h \sum_{k \in \mathbb{Z}} (\mathcal{A}p)(kh) u(x - kh), \quad x \in \mathbb{R}, \quad (11.9)$$

and, on $[-1, 1]$, compact support reduces this to

$$p(x) = h \sum_{k \in K_n} (\mathcal{A}p)(kh) u(x - kh). \quad (11.10)$$

Proof. Define the sampled synthesis operator

$$(S_h v)(x) = h \sum_{k \in \mathbb{Z}} v(kh) u(x - kh).$$

Apply Poisson summation to $t \mapsto e^{zx} u(x - t)$. For z near zero,

$$S_h(e^{z\cdot})(x) = \sum_{\ell \in \mathbb{Z}} e^{(z - 2\pi i \ell/h)x} \mathcal{M}_u(-z + 2\pi i \ell/h). \quad (11.11)$$

Because u is even, the $\ell = 0$ term is $e^{zx} \mathcal{M}_u(z)$. If $h = 2^{-n}$ and $\ell \neq 0$, then $\ell/h = 2^n \ell$; by (11.2), the corresponding factor has a zero of order at least $n + 1$ at $z = 0$. Therefore the Taylor coefficients through degree n in (11.11) coincide with those of $e^{zx} \mathcal{M}_u(z)$. Equating the coefficient of $z^r/r!$ gives

$$S_h(t^r)(x) = \mathcal{M}_u(D)x^r, \quad 0 \leq r \leq n. \quad (11.12)$$

By linearity, $S_h v = \mathcal{M}_u(D)v$ for every polynomial v of degree at most n . Taking $v = \mathcal{A}p$ proves (11.9).

For $x \in [-1, 1]$, a term can be nonzero only when $|x - kh| \leq 1$, hence $kh \in [-2, 2]$. The two extreme indices $k = \pm 2m$ contribute $u(\pm 1) = 0$, leaving precisely K_n . $\square$

The exponential relation between degree and spacing is sharp on uniform integer-ratio lattices: the first nonzero dual-lattice frequency has zero multiplicity $n + 1$ only when $1/h$ is divisible by 2^n . Thus the same 2-adic valuation that controls the Fourier product determines the cost of exact polynomial synthesis.

11.3 A q -Appell deformation of geometric falling powers

Apply $\mathcal{A}$ to the geometric Newton basis. Define

$$\mathcal{A}_k(y; c, q) := (\mathcal{A}N_k(\cdot; c, q))(y). \quad (11.13)$$

Proposition 11.2 (Elementary-symmetric formula). *For $k \geq 0$,*

$$\mathcal{A}_k(y; c, q) = \sum_{r=0}^{\lfloor k/2 \rfloor} \gamma_{2r} e_{k-2r}(y - c, y - cq, \dots, y - cq^{k-1}). \quad (11.14)$$

Consequently

$$N_k(x; c, q) = h \sum_{s \in K_n} \mathcal{A}_k(sh; c, q) u(x - sh), \quad k \leq n, \quad (11.15)$$

for $x \in [-1, 1]$ and $h = 2^{-n}$.

Proof. For a product of k linear factors,

$$\frac{1}{r!} \frac{d^r}{dy^r} \prod_{j=0}^{k-1} (y - cq^j) = e_{k-r}(y - c, y - cq, \dots, y - cq^{k-1}).$$

Substitute $r = 2s$ into (11.7), and then apply Theorem 11.1. $\square$

The polynomials $\mathcal{A}_k$ simultaneously deform two classical sequences: setting all inverse moments except γ_0 to zero recovers the q -Pochhammer Newton basis, while letting $q \rightarrow 1$ in logarithmic coordinates recovers inverse-moment Appell transforms of ordinary falling factorials. Their mixed recurrences, zero geometry, and connection coefficients appear to be unexplored.

11.4 Geometric cardinals as finite sums of up translates

Assume now $0 < c \leq 1$, so the comb nodes lie in $[-1, 1]$. Let $L_{j,n}^{c,q} := \ell_{j,n}$ be the geometric cardinal from Theorem 4.1.

Theorem 11.3 (Gaussian–Appell cardinal decoder). *For $0 \leq j \leq n$ and $x \in [-1, 1]$,*

$$L_{j,n}^{c,q}(x) = h \sum_{k \in K_n} D_{k,j}^{(n;c,q)} u(x - kh), \quad (11.16)$$

where

$$D_{k,j}^{(n;c,q)} := (\mathcal{A}L_{j,n}^{c,q})(kh) \quad (11.17)$$

$$= \lambda_{j,n}^{c,q} \sum_{r=0}^{\lfloor n/2 \rfloor} \gamma_{2r} e_{n-2r}(\{kh - cq^s : 0 \leq s \leq n, s \neq j\}), \quad (11.18)$$

with

$$\lambda_{j,n}^{c,q} = c^{-n} \frac{(-1)^j q^{-nj + \binom{j+1}{2}}}{(q; q)_j (q; q)_{n-j}}. \quad (11.19)$$

Thus every cardinal coefficient is an explicit product of a Gaussian barycentric prefactor and a finite Bernoulli–Appell symmetric polynomial.

Proof. Apply [Theorem 11.1](#) to the degree- n polynomial $L_{j,n}^{c,q}$. Since

$$L_{j,n}^{c,q}(y) = \lambda_{j,n}^{c,q} \prod_{\substack{0 \leq s \leq n \\ s \neq j}} (y - cq^s),$$

the same derivative calculation as in [Theorem 11.2](#) yields (11.18). The prefactor is the barycentric weight in (4.2). $\square$

Corollary 11.4 (Interpolation through an atomic dictionary). *For arbitrary nodal data $y_0, \dots, y_n$,*

$$(\mathcal{I}_n^{c,q} y)(x) = \sum_{k \in K_n} C_k(y) u(x - kh), \quad (11.20)$$

where

$$C_k(y) = h \sum_{j=0}^n D_{k,j}^{(n;c,q)} y_j. \quad (11.21)$$

The map from data to atom amplitudes is finite and linear.

11.5 Collocation, a stochastic encoder, and a signed right inverse

Let B be the $(n+1) \times (4 \cdot 2^n - 1)$ collocation matrix and D the $(4 \cdot 2^n - 1) \times (n+1)$ decoder matrix,

$$B_{i,k} = u(cq^i - kh), \quad D_{k,j} = D_{k,j}^{(n;c,q)}. \quad (11.22)$$

Indices k range over K_n .

Theorem 11.5 (Exact Gaussian–Appell biorthogonality). *The matrices satisfy*

$$(hB)D = I_{n+1}. \quad (11.23)$$

In components,

$$h \sum_{k \in K_n} u(cq^i - kh) (\mathcal{A} L_{j,n}^{c,q})(kh) = \delta_{ij}. \quad (11.24)$$

Moreover, hB is nonnegative and row-stochastic, while every row of D sums to 1.

Proof. Evaluate (11.16) at $x = cq^i$ to obtain (11.24). Constant reproduction in [Theorem 11.1](#) gives $h \sum_k u(x - kh) = 1$, so every row of hB sums to one; nonnegativity follows from $u \geq 0$. Finally

$$\sum_{j=0}^n D_{k,j} = \mathcal{A} \left(\sum_{j=0}^n L_{j,n}^{c,q} \right) (kh) = (\mathcal{A} 1)(kh) = 1.$$

$\square$

Thus a positive averaging map has an exact signed polynomial decoder. The right inverse cannot generally be nonnegative: reconstructing high-degree cardinals requires cancellation, and the size of that cancellation reflects the geometric Lebesgue growth.

11.6 Closing the loop for the Rvachev function

Take $c = 1$, $q = 1/2$, and interpolate u at the center-directed comb $1, 1/2, 1/4, \dots$. Since

$$u(2^{-j}) = F(1 - 2^{-j}) = 1 - F(2^{-j}), \quad (11.25)$$

all nodal values are rational and the polynomial interpolant satisfies

$$\mathcal{I}_n u = 1 - \mathcal{I}_n F. \quad (11.26)$$

Combining this identity with [Theorem 11.4](#) gives a finite self-representation.

Theorem 11.6 (Finite geometric Lagrange–Rvachev loop). *Put*

$$U_n(x) := h \sum_{k \in K_n} (\mathcal{A}(\mathcal{I}_n u))(kh) u(x - kh), \quad h = 2^{-n}. \quad (11.27)$$

Then

$$\boxed{U_n(x) = (\mathcal{I}_n u)(x), \quad -1 \leq x \leq 1.} \quad (11.28)$$

Hence, on every set and in every norm or seminorm on which both sides are defined,

$$\|u - U_n\| = \|u - \mathcal{I}_n u\|. \quad (11.29)$$

The atomic substitution creates no additional truncation error at a fixed level.

Proof. Equation (11.28) is [Theorem 11.1](#) applied to the polynomial $\mathcal{I}_n u$. The error identity follows by subtraction. $\square$

For degree 2, the center-comb interpolant is

$$(\mathcal{I}_2 u)(x) = \frac{40}{27} - \frac{22}{9}x + \frac{26}{27}x^2. \quad (11.30)$$

Since $h = 1/4$, $K_2 = \{-7, \dots, 7\}$, and $\mathcal{A}p = p - p''/18$ for a quadratic, one obtains the concrete loop

$$\boxed{\frac{40}{27} - \frac{22}{9}x + \frac{26}{27}x^2 = \sum_{k=-7}^7 \left(\frac{13k^2}{864} - \frac{11k}{72} + \frac{167}{486} \right) u\left(x - \frac{k}{4}\right),} \quad (11.31)$$

valid for $-1 \leq x \leq 1$.

The loop also transfers instability exactly. At the accumulation point $x = 0$, [Theorem 10.3](#) gives recovery of every fixed center jet of u . In the gap $(1/2, 1)$, equation (11.26) transfers the Fabius divergence to u . A compactly supported atomic representation is therefore a change of coordinates, not a cure for a badly conditioned interpolation projector.

11.7 The reflected boundary comb

The affine reflection $t = 1 - x$ sends the inverse-dyadic comb to

$$t_j = 1 - 2^{-j}, \quad j \geq 0, \quad (11.32)$$

which accumulates at the support boundary $t = 1$. The polynomial $J_n(t) = (\mathcal{I}_n F)(1 - t)$ interpolates $u(t)$ at these nodes. Its Newton basis is

$$\prod_{r=0}^{k-1} ((1 - t) - 2^{-r}), \quad (11.33)$$

and all formulas above follow by affine covariance. This is the natural comb for boundary-flat approximation, while 2^{-j} is the natural comb for center-flat approximation. The two are conjugate through the Fabius symmetry, but their geometric interpretation inside the support of u is different.

Part IV

Experiments, conjectures, and implementation

Exact and high-dynamic-range numerical experiments

The accompanying program `geometric_comb_experiments.py` was written to test formulas whose normalizations are easy to misstate and to explore asymptotic regimes beyond ordinary floating-point range. It separates exact rational calculations from logarithmic floating-point calculations.

12.1 Reproducibility protocol

The exact path uses Python's `Fraction` for:

1. the Fabius recurrence (9.9);
2. every value $F(2^{-j})$ used in the experiments;
3. direct Lagrange evaluation at $x = 0$ and $x = 3/4$;
4. Gaussian Pascal coefficients at integer base 2;
5. verification of the basis transform and the complete-homogeneous residual identity.

No tolerance enters these checks.

The Lebesgue function is positive but astronomically large. It is therefore computed from logarithms of the absolute cardinal factors, followed by a log-sum-exp reduction. A coarse scan isolates the outer boundary layer and a golden-section search refines the maximum. The analytic constants $(-q; q)_\infty / (q; q)_\infty$ and $2(-q; q)_\infty / e$ are computed with `mpmath` at elevated precision.

Running the script from its directory creates the tabular outputs `data/*.csv`, the exact-check transcript `data/identity_checks.txt`, and the figure files `figures/*.pdf` and `figures/*.png`. The PDF figures are used directly in this report.

12.2 Exact algebraic checks

The script verifies, in exact arithmetic,

$$x^m = \sum_{k=0}^m c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q N_k(x; c, q), \quad 0 \leq m \leq 9, \quad (12.1)$$

$$x^{n+1+r} - \mathcal{I}_n(x^{n+1+r}) = N_{n+1}(x; c, q) h_r(x, c, cq, \dots, cq^n), \quad 0 \leq n \leq 7, \quad 0 \leq r \leq 5. \quad (12.2)$$

The test values $q = 1/2$, $c = 3/2$, $x = 7/5$ in the first check and $q = 1/2$, $c = 4/3$, $x = 11/13$ in the second are deliberately nonsingular and nontrivial. All tested identities pass exactly.

The first thirteen Fabius Newton coefficients are also generated exactly. The first six agree with (9.17)–(9.18); the later fractions are retained in `data/identity_checks.txt` rather than typeset here.

12.3 Lebesgue growth and boundary-layer location

For $q = 1/2$ and $c = 1$, the two constants are

$$\frac{(-q; q)_\infty}{(q; q)_\infty} = 8.25598793577825006554414084943 \dots, \quad (12.3)$$

$$\frac{2(-q; q)_\infty}{e} = 1.75421915716734780836138830414 \dots \quad (12.4)$$

Selected computed values are shown in [Table 12.1](#).

Table 12.1: Geometric-comb Lebesgue diagnostics for $q = 1/2$, $c = 1$. The last column is the computed maximum divided by the leading asymptotic in (7.13).

n	$\Lambda_n(0)$	x_n^*	$n(1 - x_n^*)$	$\Lambda_n^*/\Lambda_{n,\text{asympt}}^*$
4	7.285714	0.830861	0.676556	0.86792
8	8.191739	0.899264	0.805886	0.86854
12	8.251958	0.927713	0.867445	0.90841
16	8.255736	0.943752	0.899970	0.93089
20	8.255972	0.954011	0.919782	0.94464
24	8.255987	0.961122	0.933075	0.95384
30	8.255988	0.968453	0.946418	0.96306

The endpoint column rapidly approaches its finite limit. At the same time, $\log_{10} \Lambda_n^*$ reaches approximately 129.70 by $n = 30$. The ratio in the last column and the scaled peak location both approach one, in agreement with [Theorem 7.3](#).

12.4 Exact Fabius interpolation in the gap

At $x = 3/4$ the target value is exact:

$$F(3/4) = 1 - F(1/4) = \frac{67}{72}. \quad (12.5)$$

Every interpolant value in the experiment is a rational number computed from the exact inverse-dyadic samples. The error initially fluctuates at a moderate scale and then enters a clear quadratic-logarithmic growth regime.

Table 12.2: Exact Fabius interpolation diagnostics. The third column is the number of correct decimal digits in the extrapolated boundary value $R_n = (\mathcal{I}_n F)(0)$, measured as $-\log_{10} |R_n|$.

n	$\log_{10} (\mathcal{I}_n F)(3/4) - 67/72 $	$-\log_{10} R_n $
4	-0.8013	2.1595
8	0.8750	6.7622
12	3.7620	14.3862
16	10.5655	24.8392
20	19.0624	37.9904
24	29.6591	53.8998
30	49.7229	81.9068
40	94.5437	140.6070
50	153.7151	215.2423

By $n = 50$, the same exact interpolant that predicts $F(0)$ to more than 215 decimal digits is wrong at $3/4$ by about 10^{154} . This is a particularly clean numerical manifestation of the stability dichotomy.

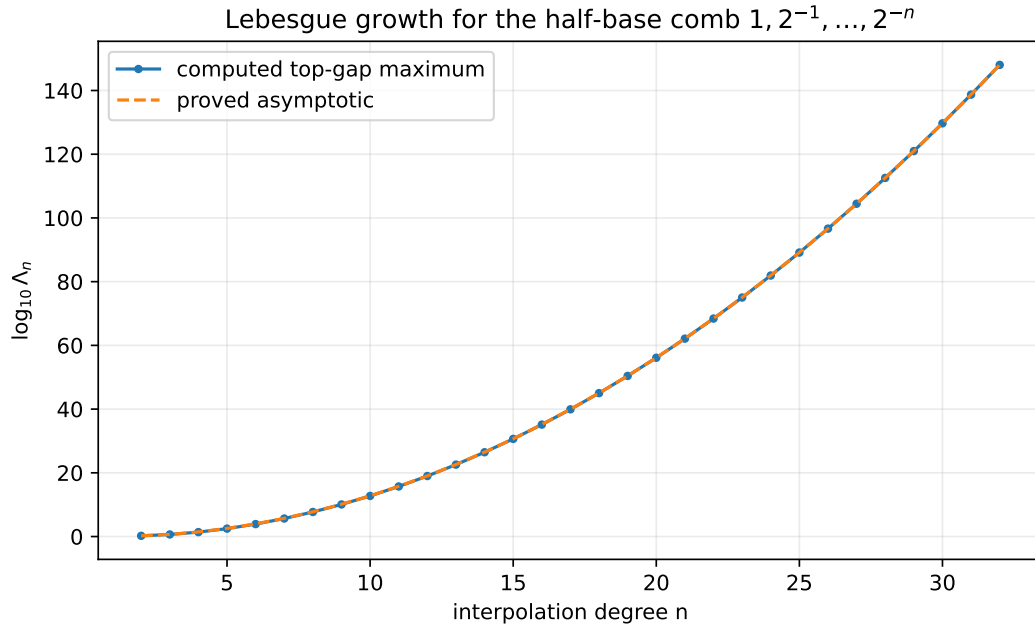


Figure 12.1: Computed global Lebesgue maxima versus the sharp asymptotic $2(-q; q)_\infty q^{-\binom{n}{2}} / (en)$ for $q = 1/2$. Logarithmic arithmetic is essential: the ordinate already exceeds 10^{148} at $n = 32$.

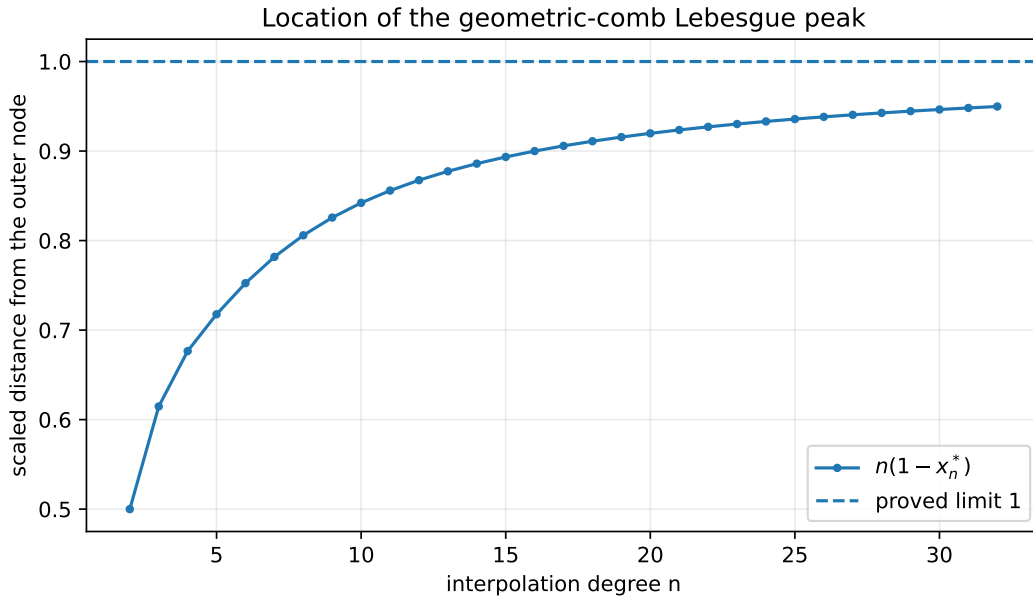


Figure 12.2: The scaled boundary-layer location $n(1 - x_n^*)$. The limiting value 1 is the maximizer of $\alpha e^{-\alpha}$ in Theorem 7.2.

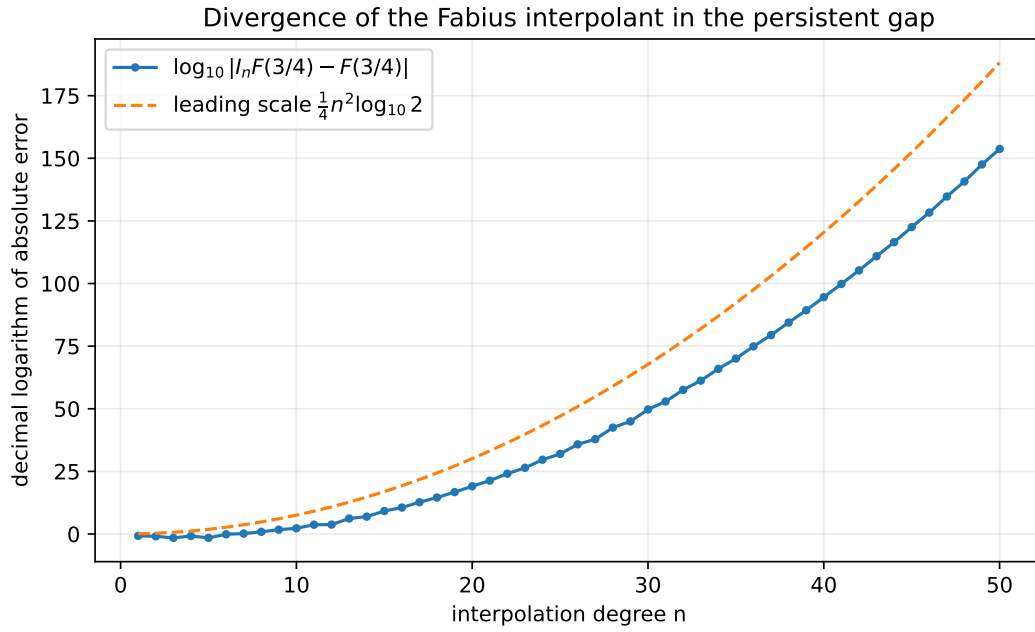


Figure 12.3: Exact-rational Fabius interpolation error at $x = 3/4$. The plotted quadratic comparison is the leading $n^2/4$ term suggested by [Theorem 10.6](#); the negative $(n/2) \log n$ correction is visible as a substantial downward displacement.

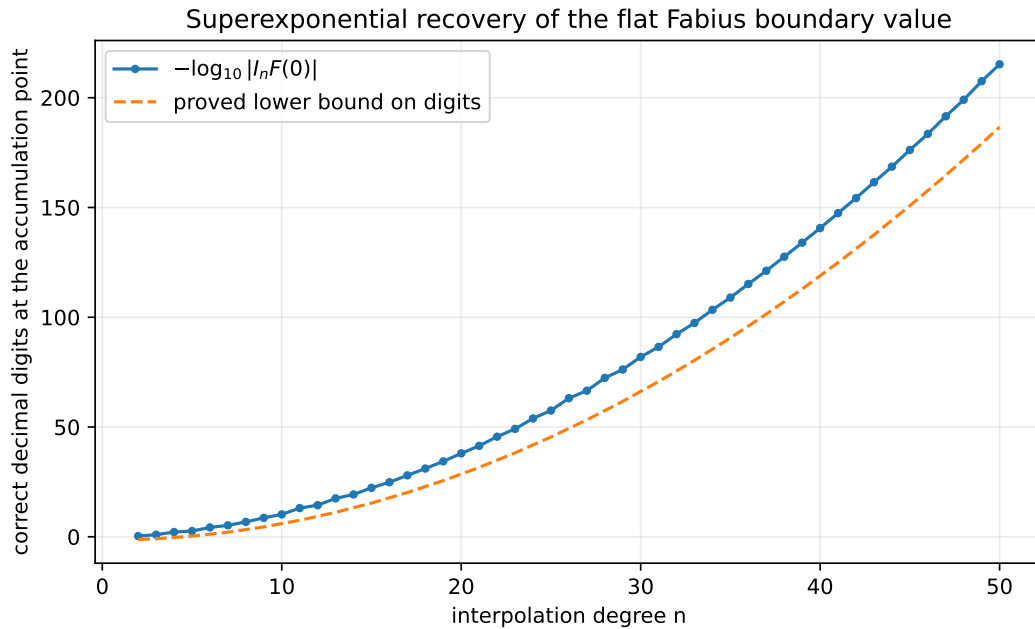


Figure 12.4: Decimal digits recovered at the accumulation point. The exact residue is compared with the rigorous quadratic-exponential estimate from [Theorem 10.2](#).

Numerical observation 12.1 (Sign oscillation). The gap error does not have a simple alternating sign. Long runs of two consecutive signs occur, reflecting the motion of a discrete saddle across integer indices. This supports the modulation clause in [Theorem 10.6](#) and argues against fitting a single smooth prefactor from small n .

12.5 What the experiments do and do not establish

The exact checks strongly guard against transcription errors in powers of q , Gaussian reciprocity, and factorial normalizations. The numerical Lebesgue data confirm the proved asymptotic but are not needed for its proof. The Fabius gap data, by contrast, are evidence for a conjecture: they show rapid divergence at one exact rational point but do not prove the proposed general saddle law. The script records raw CSV data so alternative asymptotic fits can be tested without repeating the expensive rational interpolation.

Further deductions, conjectures, and research directions

This chapter separates deductions that follow mechanically from the proved framework from genuinely open asymptotic and structural questions.

13.1 Second-order Lebesgue asymptotics

The proof of [Theorem 7.2](#) can be expanded one order further. The factors $(q/x; q)_{n-1}$, the ratio cloud (7.8), and $(1 - \alpha/n)^{n-1}$ each contribute an explicit $1/n$ correction. This suggests the following refinement.

Conjecture 13.1 (Second-order boundary layer). There exist explicit constants $A(q)$ and $B(q)$ such that

$$\frac{x_n^*}{c} = 1 - \frac{1}{n} + \frac{A(q)}{n^2} + O_q(n^{-3}), \quad (13.1)$$

$$\Lambda_n^* = \frac{2(-q; q)_\infty}{e} \frac{q^{-\binom{n}{2}}}{n} \left(1 + \frac{B(q)}{n} + O_q(n^{-2}) \right). \quad (13.2)$$

The coefficients should be expressible through Lambert series such as $\sum_{r \geq 1} q^r / (1 - q^r)$ and weighted Euler-cloud moments.

A proof should follow from a uniform expansion of the ratios of the last $O(\sqrt{\log n})$ cardinals, with the remainder controlled by the $q^{\binom{r}{2}}$ tail. The result would turn the qualitative convergence in [Table 12.1](#) into a high-accuracy formula.

13.2 A function-space criterion for infinite geometric Newton series

For analytic targets, [Theorem 6.1](#) supplies a convenient sufficient condition. A more intrinsic question starts from a data sequence $y_j = f(cq^j)$ and asks whether the formal Newton series

$$\sum_{k \geq 0} \left(c^{-k} \sum_{j=0}^k \frac{(-1)^j q^{-jk + \binom{j+1}{2}}}{(q; q)_j (q; q)_{k-j}} y_j \right) N_k(x; c, q) \quad (13.3)$$

converges in a prescribed domain.

Research problem 13.2 (Geometric Newton sequence spaces). Characterize weighted sequence spaces Y on the data $y = (y_j)$ and coefficient spaces G on the divided differences $\gamma = (\gamma_k)$ for which the triangular Gaussian transform is an isomorphism and the Newton synthesis map is bounded into $C(K)$, a Hardy space, or a Bergman space. Determine the exact dependence of the norm on the location of K relative to the isolated outer node.

The Richardson generating factorization (4.11) indicates that Euler products provide the correct multipliers on the data side. A Wiener-algebra formulation may make inversion and perturbation estimates especially transparent.

13.3 Signed saddle analysis for Fabius data

The proof of Theorem 10.5 isolates the unsigned saddle at (10.15). To prove Theorem 10.6, one needs asymptotics for the signed local packet

$$\sum_{|j-j_n| \leq C\sqrt{\log n}} (-1)^j \frac{2^{j(n-j)} a_j}{x - 2^{-j}} \quad (13.4)$$

with relative error smaller than the packet itself.

A promising route has three stages:

1. derive uniform saddle asymptotics for $a_j = [z^j]A(z)$ from the product (9.7);
2. insert them into (13.4) and apply a discrete Gaussian or theta transformation;
3. retain the fractional saddle displacement, which should produce a bounded periodic or antiperiodic modulation.

The functional equation (9.8) suggests that the moment saddle itself carries a logarithmically periodic correction. Its interaction with the integer Gaussian saddle may be the source of the sign blocks seen in the experiment.

Research problem 13.3 (Exceptional gap points). Determine whether there are points $x \in (1/2, 1)$ at which the leading signed saddle cancels identically or along an infinite subsequence. If such points exist, classify them arithmetically and determine the next asymptotic scale.

13.4 Boundary-residue entire functions

The generating function

$$\mathcal{R}(z) = (qz; q)_\infty \sum_{j \geq 0} \frac{q^{\binom{j}{2}} a_j}{(q; q)_j} z^j \quad (13.5)$$

encodes all endpoint extrapolation errors. The Euler factor has prescribed zeros at $z = q^{-r}$, while the second factor is an entire transform of the Fabius moments.

Research problem 13.4 (Zero cancellation and coefficient asymptotics). Determine which Euler zeros survive in $\mathcal{R}$, obtain the order and indicator of $\mathcal{R}$, and use its zero distribution to derive a sharp asymptotic for R_n . In particular, decide whether

$$\log |R_n| = -\frac{\log 2}{4} n^2 + O(n \log n) \quad (13.6)$$

has a full expansion with a parity-dependent linear term.

The exact coefficient formula makes this problem accessible both to complex-analytic saddle methods and to 2-adic valuation experiments.

13.5 Geometric Hermite stability

The anchored factorization (8.9) removes confluent linear algebra but not operator growth. For a flat target, its data are rescaled by $(cq^j)^{-(m+1)}$, changing the Gaussian saddle. A first model replaces a_j by $q^{\binom{j}{2}-j(m+1)}a_j$ in the Fabius formulas. This predicts a movable saddle and a phase transition when m is allowed to grow with n .

Conjecture 13.5 (Hermite saddle displacement). For $q = 1/2$ and Fabius data, the dominant index in the gap for the m -anchored interpolant is

$$j_{n,m} = \frac{n}{2} + m + 1 - \frac{\log n}{2 \log 2} + O(1), \quad (13.7)$$

until it collides with the endpoint $j = n$. The collision marks a change from an interior Gaussian saddle to an endpoint saddle and should determine the largest useful jet order.

This conjecture follows from the unsigned ratio heuristic; the signed problem may shift the practical optimum. It can be tested by extending the bundled exact code to [Theorem 8.4](#).

13.6 The mixed q -Appell family

The polynomials $\mathcal{A}_k(y; c, q)$ in (11.14) deserve study independently of interpolation. They combine a Sheffer/Appell deconvolution with a q -falling-factorial basis.

Research problem 13.6 (Structure of the q -Appell falling powers). Find:

1. raising and lowering operators for $\mathcal{A}_k$ involving D_q and the ordinary derivative D ;
2. their connection coefficients with monomials, ordinary Appell polynomials, and N_k ;
3. generating functions in k and continued-fraction representations;
4. zero-location and interlacing theorems for $0 < q < 1$;
5. asymptotics in the joint limits $k \rightarrow \infty$, $q \rightarrow 1$, and $k(1 - q) \rightarrow \tau$.

A useful starting identity is

$$\sum_{k \geq 0} \mathcal{A}_k(y; c, q) \frac{t^k}{(q; q)_k} = \mathcal{M}_u(D_y)^{-1} \sum_{k \geq 0} N_k(y; c, q) \frac{t^k}{(q; q)_k}, \quad (13.8)$$

where the second series can be rewritten by the infinite q -binomial theorem after normalization. Making the operator interchange analytic is itself a worthwhile problem.

13.7 Decoder variation, total positivity, and optimal gauges

The matrix hB in [Theorem 11.5](#) is a positive row-stochastic sampling operator. Its right inverse D has row sum one but must use signs. The size of its row variation controls amplification from nodal perturbations to atom amplitudes.

Research problem 13.7 (Minimum-variation atomic decoder). Among all right inverses of the collocation matrix hB , minimize a weighted ℓ^1 or total-variation norm subject to exact polynomial reproduction. Compare the canonical Appell decoder with the minimizer, and determine whether a symmetry-preserving minimizer has a closed q -binomial formula.

The dense dictionary has an alias kernel inherited from the dyadic Fourier zeros. Adding an alias vector changes amplitudes without changing the synthesized polynomial. Convex optimization over this affine space may substantially improve numerical conditioning while preserving every exact identity.

13.8 Stable alternatives to global gap interpolation

The superexponential Lebesgue growth is a property of the all-node global projector, not of geometric data in general. Several alternatives retain the dyadic arithmetic:

1. piecewise low-degree Newton or Hermite interpolation on geometric annuli $[cq^{j+1}, cq^j]$;
2. rational barycentric interpolation with fixed local degree;
3. oversampled least squares in a logarithmic coordinate;
4. direct approximation in scaled translates of u rather than first passing through one global polynomial;
5. hybrid Taylor–Newton schemes that use the flat germ near zero and a separate approximation across the outer gap.

For the Fabius function, the functional equation can transport a local approximation between adjacent dyadic scales, suggesting multiresolution algorithms with exact consistency constraints.

13.9 Complex and arithmetic geometric combs

When q is complex with $0 < |q| < 1$, the comb spirals into the origin. The algebraic formulas remain valid, while the stability problem becomes potential-theoretic and angle-dependent. At a root of unity the nodes collide and ordinary interpolation becomes confluent; the denominator-free Gaussian recurrence is then the correct algebraic replacement.

Over finite fields, the condition that $q^0, \dots, q^n$ be distinct is exactly an order condition in the multiplicative group. The geometric Vandermonde determinant in [Theorem 2.5](#) records its failure cyclotomically. This suggests a finite-field interpolation theory whose transition matrices are literal Gaussian coefficients counting subspaces.

Research problem 13.8 (Cyclotomic confluent limit). Develop a division-free specialization of the cardinal and residual identities when q approaches a primitive root of unity. Identify the resulting Hermite blocks, their cyclotomic valuations, and their relation to q -Lucas congruences.

13.10 Formalization program

The theory separates into finite algebra, analytic convergence, and asymptotics, making it well suited to staged formalization in Lean.

1. **Finite products.** Define N_k , prove (2.2), and derive the denominator product in [Theorem 4.1](#) without division until the final field specialization.
2. **Gaussian Pascal equivalence.** Formalize [Theorem 3.1](#) and its inverse as mutually inverse triangular matrices over a commutative ring.
3. **Residual calculus.** Prove [Theorem 5.1](#) through generating functions for complete homogeneous symmetric polynomials, then derive the analytic residual only after importing convergence lemmas.
4. **Endpoint stability.** Formalize the exact identity (7.2) and monotone convergence of finite q -products.
5. **Boundary profile.** Isolate the elementary limits $(1 - \alpha/n)^n \rightarrow e^{-\alpha}$, dominated convergence for the Euler cloud, and the tail bound from $q^{\binom{r}{2}}$.

6. **Fabius arithmetic.** Reuse the existing exact dyadic-value development to prove [Theorem 9.3](#), the residue generating function, and the fixed-jet bound.
7. **Atomic synthesis.** Formalize the finite consequence (11.16) once the repository's polynomial reproduction theorem for u_p is available; matrix biorthogonality then reduces to finite sums.

The sharp global Lebesgue theorem is the most analytically demanding new formal target. Its proof was deliberately organized into local profile, away-from-boundary decay, and compactness of the scaled maximizer so these pieces can be formalized independently.

Formula dictionary

This appendix collects the principal transforms in one place. It is meant as a reference sheet rather than a substitute for the proofs.

A.1 General geometric comb

For $x_j = cq^j$, $0 < q < 1$, define

$$N_k(x) = \prod_{r=0}^{k-1} (x - cq^r), \quad \omega_{n+1}(x) = N_{n+1}(x).$$

Then

Table A.1: Core geometric-comb formulas.

Object	Formula
Newton/ q -Pochhammer basis	$N_k(x) = (-c)^k q^{\binom{k}{2}} (x/c; q^{-1})_k$
Divided difference	$[f; c, \dots, cq^k] = D_q^k f(c) / [k]_q!$
Explicit data transform	$c^{-k} \sum_{j=0}^k (-1)^j q^{-jk + \binom{j+1}{2}} f(cq^j) / ((q; q)_j (q; q)_{k-j})$
Monomial to Newton	$x^m = \sum_{k=0}^m c^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q N_k(x)$
Newton to monomial	$N_k(x) = \sum_{m=0}^k (-1)^{k-m} c^{k-m} q^{\binom{k-m}{2}} \begin{bmatrix} k \\ m \end{bmatrix}_q x^m$
Barycentric weight	$\lambda_{j,n} = c^{-n} (-1)^j q^{\binom{j+1}{2} - nj} / ((q; q)_j (q; q)_{n-j})$
Accumulation weight	$\mu_{j,n} = (-1)^{n-j} q^{\binom{n-j+1}{2}} / ((q; q)_j (q; q)_{n-j})$
Endpoint Lebesgue constant	$\Lambda_n(0) = (-q; q)_n / (q; q)_n$
Global Lebesgue asymptotic	$\Lambda_n^* \sim 2(-q; q)_\infty q^{-\binom{n}{2}} / (en)$
Monomial residual	$x^{n+1+r} - \mathcal{I}_n x^{n+1+r} = N_{n+1}(x) h_r(x, c, cq, \dots, cq^n)$
Geometric specialization	$h_r(c, cq, \dots, cq^n) = c^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q$
Jackson moment	$\int_0^c N_k(x) \, d_q x = (-1)^k c^{k+1} q^{\binom{k+1}{2}} / [k+1]_q$

A.2 Half-base Fabius formulas

For $q = 1/2$, $d_n = \mathbb{E}X^n$, and $a_n = d_n/n!$,

Table A.2: Fabius and Rvachev specialization.

Object	Formula
Moment recurrence	$(2^n - 1)a_n = \sum_{j < n} a_j / (n - j + 1)!$
Rvachev half-moment	$d_n = n \int_0^1 t^{n-1} \text{up}(t) \, dt$
Inverse-dyadic value	$F(2^{-n}) = 2^{-\binom{n}{2}} a_n$
Newton coefficient	$\nu_k = (q; q)_k^{-1} \sum_{j=0}^k (-1)^j \begin{bmatrix} k \\ j \end{bmatrix}_{q^{-1}} a_j$
Finite interpolant	$\mathcal{I}_n F(x) = \omega_{n+1}(x) (q; q)_n^{-1} \sum_{j=0}^n (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}} a_j / (x - q^j)$
Boundary residue generating function	$\sum_{n \geq 0} R_n z^n = (qz; q)_\infty \sum_{j \geq 0} q^{\binom{j}{2}} a_j z^j / (q; q)_j$
Boundary bound	$ R_n \leq e(q; q)_\infty^{-2} q^{\lfloor n^2/4 \rfloor}$
Fixed-jet bound	$ (\mathcal{I}_n F)^{(m)}(0) \leq em! \binom{n}{m} (q; q)_\infty^{-2} q^{\lfloor n^2/4 \rfloor - mn + \binom{m}{2}}$
Rvachev relation	$\text{up}(t) = F(1 - t)$ and $\mathcal{I}_n \text{up} = 1 - \mathcal{I}_n F$ on the center-directed positive comb

A.3 Atomic synthesis formulas

For a degree bound n , $h = 2^{-n}$ and $K_n = \{-2^{n+1} + 1, \dots, 2^{n+1} - 1\}$,

$$\mathcal{A}p = \mathcal{M}_u(D)^{-1}p = \sum_{r \geq 0} \frac{\gamma_{2r}}{(2r)!} p^{(2r)}, \quad (\text{A.1})$$

$$p(x) = h \sum_{k \in K_n} (\mathcal{A}p)(kh) u(x - kh), \quad \deg p \leq n, \quad (\text{A.2})$$

$$\ell_{j,n}(x) = h \sum_{k \in K_n} (\mathcal{A}\ell_{j,n})(kh) u(x - kh), \quad (\text{A.3})$$

$$(hB)D = I_{n+1}. \quad (\text{A.4})$$

Selected exact dyadic data

B.1 Moments and inverse-dyadic values

Table B.1: Regularized moments $a_n = d_n/n!$ and the corresponding values $F(2^{-n}) = 2^{-\binom{n}{2}}a_n$.

n	a_n	$F(2^{-n})$
0	1	1
1	1/2	1/2
2	5/36	5/72
3	1/36	1/288
4	143/32400	143/2073600
5	19/32400	19/33177600
6	1153/17146080	1153/561842749440
7	583/85730400	583/179789679820800
8	1616353/2623350240000	1616353/704200217922109440000

B.2 First geometric Newton coefficients

For

$$\mathcal{I}_n F(x) = \sum_{k=0}^n \nu_k \prod_{r=0}^{k-1} (x - 2^{-r}),$$

the exact coefficients begin

Table B.2: First exact half-base Newton coefficients.

k	ν_k
0	1
1	1
2	-26/27
3	-128/27
4	-4253248/637875
5	189673472/19774125
6	134859284873216/1648153544625
7	21867230161534976/209315500167375
8	-349286360632157831954432/204161105975753390625

Large numerator and denominator sizes are not numerical artifacts; they are exact consequences of composing the Fabius moment recurrence with the reciprocal Gaussian transform.

Archive and reproduction notes

C.1 Archive layout

The delivered ZIP archive contains:

Path	Purpose
<code>geometric_comb_q_fabius_repo</code>	Complete LaTeX source of this report.
<code>rt.tex</code>	
<code>geometric_comb_q_fabius_repo</code>	Rendered report.
<code>rt.pdf</code>	
<code>geometric_comb_experiments.py</code>	Exact and logarithmic numerical experiments, with detailed comments.
<code>data/lebesgue_data.csv</code>	Lebesgue endpoint values, maxima, peak locations, and asymptotic ratios.
<code>data/fabius_interpolation_data.csv</code>	Exact-rational gap and boundary diagnostics converted to plotting fields.
<code>data/identity_checks.txt</code>	Exact verification summary and Newton coefficients.
<code>figures/*.pdf, figures/*.png</code>	Reproducible figures in vector and raster form.
<code>README.md</code>	Dependencies and commands.

C.2 Commands

From the extracted archive directory, regenerate the data and figures with

```
python geometric_comb_experiments.py
```

The script requires Python 3.10 or later together with `numpy`, `matplotlib`, and `mpmath`. Rebuild the PDF with a sufficiently complete TeX Live installation, for example

```
pdflatex geometric_comb_q_fabius_report.tex
pdflatex geometric_comb_q_fabius_report.tex
```

Two passes resolve the table of contents and cross-references.

C.3 Source boundary

The primary source monograph and the neighboring reports were read to retain notation and avoid duplicating already consolidated results. The repository snapshot used here is

$$e5175d5eb78d66f4e31db3bc506541b9bae12c57 \quad (30 \text{ August } 2026). \quad (\text{C.1})$$

The sharp geometric-comb Lebesgue asymptotic, the fixed-jet Fabius estimate, and their systematic composition with the geometric Gaussian–Appell decoder are presented as new deductions relative to that snapshot.

Status ledger

Table D.1: Logical status of the principal claims.

Claim	Status	Location
Newton basis equals a normalized q -Pochhammer symbol	proved	Chapter 2
Iterated D_q equals geometric divided difference	proved	Theorem 2.2
Gaussian Pascal basis transform and inverse	proved	Theorem 3.1
Complete-homogeneous monomial residual	proved	Theorem 5.2
Analytic infinite Newton convergence	proved	Theorem 6.1
Finite endpoint Lebesgue limit	proved	Theorem 7.1
Boundary-layer profile and sharp global Lebesgue constant	proved	Theorems 7.2 and 7.3
Newton–Jackson moment formula	proved	Theorem 8.1
Fabius half-base Gaussian transform	proved	Theorem 9.3
Fabius boundary residue and fixed-jet bounds	proved	Theorems 10.1 and 10.3
Quadratic-logarithmic Fabius gap upper bound	proved	Theorem 10.5
Two-sided Fabius gap asymptotic	conjectural	Theorem 10.6
Second-order Lebesgue expansion	conjectural	Theorem 13.1
Dyadic fixed-radius polynomial synthesis by up Fourier multiplicities	proved from zero	Theorem 11.1
Gaussian–Appell cardinal decoder and right inverse	proved	Theorems 11.3 and 11.5
Optimal growing-order geometric Hermite anchoring	open	Theorem 8.6

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